# The Legibility of Income Risk*

Sven van Holten Charria†

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**Abstract**

What does the composition of pay tell workers about their earnings risk? Lower-income workers perceive greater and more persistent risk despite similar measured earnings variation across incomes. Bonuses account for more temporary variation and recur more regularly at higher incomes. In a calibrated learning model, recognising bonuses generates a modest decline in mean perceived risk with income. Knowing payment dates reproduces about half the persistence gap as annual-payment uncertainty enters and leaves successive forecasts, without requiring faster learning. Transfer experiments show that unidentified annual payments raise perceived risk beyond receipt, with much smaller effects when the same entitlement is paid monthly.

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†Pre-doctoral Research Assistant at the Centre for Macroeconomics, London School of Economics. E-mail: s.m.van-holten-charria@lse.ac.uk.

# 1 Introduction

What do the amount and composition of pay tell workers about their future earnings? Households decide how much to consume and save partly by assessing the possibility of future earnings losses. Models of household saving are commonly calibrated to earnings processes estimated from panel data, but workers may know more about their payments than those data reveal. They may anticipate some changes in pay or recognise why others will not last, using this information to assess how much uncertainty remains. A model can therefore match observed earnings variation and still misrepresent households' uncertainty and their incentive to save.

I study whether information about pay composition can help explain differences in perceived earnings risk across incomes. I compare earnings variation in the Survey of Income and Program Participation (SIPP) with workers' own assessments of future earnings risk in the New York Fed's Survey of Consumer Expectations (SCE). Lower-income workers in the SCE perceive greater risk over the coming year, conditional on keeping the same job, and their assessments are more persistent from one month to the next. The SIPP estimates do not show the same pattern: twelve-month earnings-change variances are approximately equal across income groups, and a proxy for earnings volatility is more persistent at higher incomes.

The SIPP's separate payment records reveal differences in what lies behind this earnings variation. Bonuses account for approximately 36 percent of transitory earnings variation at lower incomes and 68 percent at higher incomes. These payments generally disappear in the following month, but their annual recurrence is stronger at higher incomes. Identifying them provides information beyond that contained in total earnings histories: adding bonus shares to forecasts based on current and past total pay reduces three-month mean squared forecast error by 3.82 percent at higher incomes, compared with 0.40 percent at lower incomes. Most of the improvement comes from knowing the current bonus share, and the gains diminish at longer horizons. Higher-income workers therefore receive more variation through payment categories whose identification helps predict subsequent earnings.

I develop a Bayesian learning model to examine how workers use this information to assess earnings risk. Workers infer both permanent pay and the variability of permanent and temporary earnings from successive payments. Holding their earnings histories fixed, I compare beliefs formed from total pay alone with those formed using additional information about bonus amounts and recurring payment dates. The analytical benchmarks show how recognising temporary payments can reduce uncertainty about permanent pay and speed the correction of earlier estimation errors. To assess these effects quantitatively, I calibrate the earnings process to the SIPP and allow pay composition to change as workers move across the income distribution. Workers begin with common parameter priors and use the same learning rule, so differences in their assessments emerge from their payment histories and information. The SCE persistence estimates guide the choice of a common learning history, leaving mean-risk ratios across income groups untargeted.

The calibrated model reproduces the direction of both income gradients, but only part of their magnitude. Bonus recognition raises the lower–higher mean-risk ratio from 0.95 to 1.11, compared with 1.81 in the SCE. Higher-income workers experience more bonus variation, so identifying these payments reduces the variation they attribute to permanent shocks more strongly. Expected

permanent shocks nevertheless dominate uncertainty over the twelve-month forecast horizon, and their calibrated variances differ little across incomes, limiting the resulting differences in perceived risk. Knowing payment dates does change how these assessments evolve. A worker forecasting earnings in her annual payment month includes uncertainty about the next bonus; when her forecast moves to the following month, that contribution disappears even if her assessment of bonus variability is unchanged. Annual payments are more common and more variable at higher incomes, making these temporary increases a larger source of variation in their perceived risk. The benchmark generates a lower–higher persistence gap of 0.10, compared with 0.20 in the SCE, without requiring higher-income workers to learn faster about their earnings process.

Temporary-payment experiments show how these inferences affect responses to additional income. An unidentified one-month pay increase raises perceived risk beyond the month of receipt because the worker interprets the increase and subsequent fall as evidence of greater permanent earnings variability. Identifying the payment as a bonus removes the large positive response. In a policy application, I compare an annual government transfer with monthly instalments of the same entitlement. Monthly delivery produces much smaller risk revisions when transfers are not distinguished from earnings. When each government receipt is identified, workers remove it before updating their earnings beliefs, and neither delivery schedule changes perceived earnings risk in the model. These results favour clearly identifying government transfers and suggest that spreading payments can limit risk revisions when recipients cannot distinguish them from earnings.

These findings connect information about pay composition to the beliefs that guide consumption and saving. Differences in perceived risk can emerge from workers' payment histories even when they begin with the same beliefs, and a temporary receipt can change their desired protection against future earnings losses as well as their available resources. Accounting for this information may therefore change how consumption models distribute precautionary saving motives across households and predict spending responses to transfers. Quantifying those effects requires households' saving decisions and borrowing constraints to be modelled explicitly.

### 1.1 Related Literature

Recent research shows that measured earnings variation and the risk households perceive can differ substantially. Ben-David et al. (2018) document greater uncertainty about income growth among lower-income households. Caplin et al. (2026) and Tao Wang (2026) find that subjective risk is substantially below conventional estimates and examine the implications for saving. The former also show that job transitions account for much of subjective earnings risk. I complement this evidence by extending the comparison across incomes from the level of perceived risk to its persistence and examining whether differences in the information workers receive about their pay components can explain both patterns.

Understanding these differences requires distinguishing the sources of earnings variation. Lemieux, MacLeod and Parent (2009) show that performance pay contributes to rising wage inequality, particularly at the top of the distribution. Grigsby, Hurst and Yildirmaz (2021) show that bonuses and overtime make total compensation more flexible than base wages. Closest to my empirical analysis, Ganong et al. (2025) document frequent fluctuations in hours among

hourly workers and temporary, often predictable bonus payments among salaried workers. I build on this evidence using the SIPP's separately reported payment categories to measure their contributions to transitory variation in earnings per hour within jobs. Beyond documenting their predictability, I test whether identifying a payment's components improves forecasts relative to observing only total earnings history.

In a similar stream, variable compensation connects the paper to research on risk sharing between firms and workers. Di Maggio et al. (2022) find that higher firm uncertainty reduces compensation primarily through bonuses and commissions. Sockin and Sockin (2025) show that variable pay exposes workers to adverse firm shocks. By holding payment histories fixed and varying what workers know about their composition, I isolate the informational role of variable pay: recognising a temporary payment, even if unexpected, can help a worker infer her permanent pay without changing her underlying exposure to earnings shocks.

The model builds on the income-learning framework of Druedahl and Jørgensen (2020), in which households distinguish persistent from transitory income shocks using earnings histories and private information. They infer almost perfect information about these components from annual income and consumption data. My model also allows workers to learn earnings variances, so recognising a current payment need not resolve their uncertainty about future risk. In a simpler model, Wang (2004) shows how uncertainty about income components can generate additional precautionary saving. This also connects to Orlik and Veldkamp (2024), who show that learning about an unknown distribution, particularly its tail probabilities, generates fluctuations in subjective uncertainty. I examine how information about temporary pay changes workers' learning about permanent and temporary earnings variances under a common updating rule.

Lastly, in the application to government transfers, I focus on households' information about the stability of these payments. Sahm, Shapiro and Slemrod (2012) find a higher reported propensity to spend stimulus delivered as a lump sum than through reduced tax withholding. Thakral and Tô (2025) show that households spend more after receipt when they have had less time to anticipate a transfer. These findings motivate examining what payment design tells households about the amount, duration and recurrence of future transfers. I examine how transfer recognition and delivery frequency affect perceived earnings risk, providing a potential channel through which payment design can influence precautionary saving and households' propensity to consume out of transfers.

## 2 Data and Measurement

The empirical analysis draws on two household surveys. I use the New York Fed's Survey of Consumer Expectations (SCE) to measure workers' uncertainty about their earnings over the coming year. The Survey of Income and Program Participation (SIPP) provides monthly earnings records, including separate amounts for bonuses, commissions, tips and overtime. I describe the two datasets and outline how perceived earnings risk and realised earnings variation differ across income groups, both in their levels and persistence.

### 2.1 Data on Earnings Expectations

The SCE density question asks employed respondents for the distribution of their own earnings growth over the next twelve months, assuming that they remain in the same job, with the same employer and working the same hours.[1] In each monthly interview, respondents assign probabilities to ten ranges of possible earnings growth. Their answers form a ten-bin histogram, following the density-forecast approach of Engelberg et al. (2009). As in Wang (2026), I measure perceived earnings risk by the variance of the elicited earnings distribution. I calculate this variance using bin midpoints.[2]

The advantage of this question is that it asks workers to assess uncertainty about their pay while holding their job and hours fixed. Although I cannot verify how fully respondents follow these instructions, the elicited object is therefore closer to the conditional variance of the stochastic component of wage growth than to total income risk. In particular, it excludes the risk of losing or changing jobs, which Caplin et al. (2026) identify as an important source of subjective earnings risk. The analysis here therefore focuses on uncertainty about pay within the current job.

I restrict the sample to employees and require complete, valid histograms. I exclude responses that place all probability in a single bin and trim perceived variance at the pooled first and ninety-ninth percentiles. To follow workers' expectations within the same job, I drop each respondent's reports from the first interview in which she records a job change onwards and require at least six consecutive monthly reports. The resulting sample contains 35,697 reports from 3,790 respondents between August 2013 and February 2025, with a median of ten monthly reports per respondent. I apply survey weights throughout.

Because the two surveys interview different workers, I compare the risk reported by each income group in the SCE with earnings variation among workers in the corresponding SIPP group. I divide respondents into three groups by annual household income: below \$50,000, \$50,000–\$99,999, and at least \$100,000, which I refer to as lower, middle and higher income. The thresholds are fixed in nominal dollars over the sample. The SCE groups contain 6,746, 12,846 and 16,105 reports, respectively.

To account for differences in demographics and interview dates across the income groups, I regress perceived variance on age, sex, education, state and calendar year-month effects. The regression also controls for the number of consecutive interviews completed, since repeated participation may affect reported beliefs, as Kim and Binder (2023) document for similar questions about inflation expectations in the SCE.[3] I denote the resulting adjusted measure of perceived earnings risk by $\widetilde{v}_{it}$.

[1] The SCE question on earnings uncertainty (Q24) asks: *"Suppose again that, 12 months from now, you are working in the exact same job at the same place you currently work, and working the exact same number of hours. In your view, what would you say is the percent chance that 12 months from now..."* It continues: *"Your earnings on this job, before taxes and deductions, will have..."* For respondents with multiple jobs, the question inserts "main" before "job". The ten response options cover increases and decreases of 0–2%, 2–4%, 4–8%, 8–12%, and 12% or more.

[2] I assign each bin's probability to its midpoint, using ±1, ±3, ±6 and ±10 percent for the closed intervals and ±16 percent for the open-ended tails, following the medium-width specification of Krüger and Pavlova (2024).

[3] Age enters linearly and quadratically. The interview controls distinguish the first, second, and subsequent interviews within each uninterrupted sequence of monthly responses, with the twelfth and all later interviews grouped together.

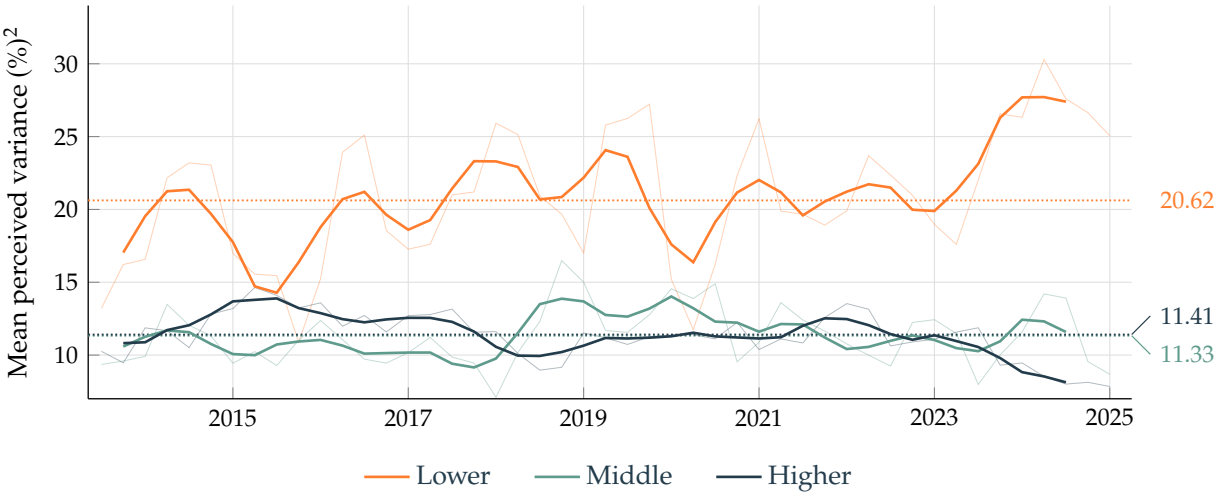



**Figure 1:** Perceived earnings variance by household-income group

*Notes:* SCE adjusted perceived variance, residualised, in squared percentage points. Thin lines show survey-weighted quarterly means; thick lines show their four-quarter centred moving averages. Dotted lines mark the survey-weighted means over the full sample. Lower-income workers report about 1.8 times as much perceived variance as higher-income workers.

Figure 1 shows that lower-income workers report greater earnings uncertainty, while the middle- and higher-income groups track each other closely. The lower–higher gap is present before and after adjusting for demographics, interview timing and survey participation, and also survives tighter income cut-offs (see Appendix A).

### 2.2 Data on Earnings and Pay Components

The SIPP is a longitudinal household survey conducted by the U.S. Census Bureau. Annual interviews collect employment and income histories for each month of the preceding calendar year. Alongside total earnings, the survey records pay rates, hours worked, and separate amounts for bonuses, commissions, tips and overtime. These details allow me to examine both changes in a worker's total pay and the payments that account for them.

I use earnings from the first recorded job, excluding self-employment and retaining uninterrupted periods of at least six months in the same job. I require positive earnings and reported hours. The underlying component panel contains 2,250,815 monthly observations for 103,134 workers over 2013–2023, with a median of seventeen observed months per worker.

I separate monthly earnings into five components: the four itemised payments and the wages and salaries that remain after subtracting them from total earnings. Table 1 shows that wages and salaries excluding itemised pay account for about 96 percent of annual earnings in each income group. The mean annual bonus share rises sharply with income, from 0.26 to 1.24 percent, while the share from tips falls. This pattern is consistent with differences in the types of jobs held across income groups, with bonuses associated with professional and managerial roles and tips with service work.

A payment can account for a small share of annual earnings and still cause a large monthly change if it is concentrated in a single month. Among workers observed in the same job for a complete

calendar year, 3.2 percent of the lower-income group receive a bonus, compared with 11.2 percent of the higher-income group. Bonuses are also larger at higher incomes: in months when a bonus is received, the mean payment rises from $985 to $10,150, or from 22.5 to 30.4 percent of monthly pay. These means conceal substantial dispersion. In the higher-income group, for example, the median positive bonus is $2,030. Appendix Table B.1 reports the long upper tail of bonus payments and the corresponding distributions of the other components.

**Table 1:** Annual earnings shares, receipt and size of pay components

| | Lower | Middle | Higher |
|---|---|---|---|
| *A. Mean share of annual earnings (%)* | | | |
| Wages and salaries excluding itemised pay | 96.65 | 96.52 | 95.80 |
| Bonuses | 0.26 | 0.44 | 1.24 |
| Commissions | 1.88 | 1.90 | 2.06 |
| Tips | 0.88 | 0.53 | 0.35 |
| Overtime | 0.33 | 0.61 | 0.56 |
| *B. At least one payment during the year (%)* | | | |
| Bonuses | 3.2 | 6.0 | 11.2 |
| Commissions | 2.6 | 3.2 | 3.9 |
| Tips | 2.5 | 1.5 | 1.0 |
| Overtime | 4.5 | 6.2 | 5.2 |
| *C. Mean monthly payment, conditional on receipt ($)* | | | |
| Bonuses | 985 | 1,978 | 10,150 |
| Commissions | 1,881 | 2,694 | 6,753 |
| Tips | 674 | 935 | 1,077 |
| Overtime | 396 | 708 | 1,228 |

*Notes:* Survey-weighted estimates, 2013–2023; amounts in nominal dollars. Panels A and B use complete same-job years; C conditions on receipt. Panel A averages workers' annual shares, including zeros. Hours may vary.

Monthly earnings can change because a worker's pay changes or because she works more or fewer hours. Ganong et al. (2025) show that fluctuations in hours account for much of monthly earnings instability among hourly workers. To separate these sources of variation, I divide total monthly pay by reported monthly hours and study changes in hourly earnings. This measure includes the four additional payment categories, so a bonus can raise earnings per hour even when the worker's contractual pay rate is unchanged.

For the comparison with perceived risk, I retain months in which reported hours change. Requiring unchanged hours would exclude a substantial share of earnings variation, particularly among lower-income workers, whose hours vary more (Appendix Figure C.1). These payments may still inform workers' assessments of future pay, even when they are asked to hold future hours fixed. I therefore use the full same-job histories and report the unchanged-hours comparison in Appendix C.1.

Before measuring earnings variation, I regress log hourly earnings on age, age squared, sex, race and education, with state and calendar year-month effects. I then study monthly changes in the residuals. For the variance decomposition, I exclude December-to-January changes because the two months are reported in separate annual interviews, creating a well-documented survey seam that produces artificial jumps in recorded earnings. I also trim squared residual changes at the

pooled first and ninety-ninth percentiles. These adjustments do not ensure that the remaining changes are unexpected. For instance, a worker who knows that she will receive a scheduled pay rise faces no uncertainty about that increase, yet it can remain in residual earnings growth if the controls do not capture its timing.

### 2.3 Connecting the SCE and SIPP

To relate monthly earnings variation in the SIPP to workers' reported uncertainty in the SCE, I use a simple version of Meghir and Pistaferri's (2004) permanent–transitory framework. It separates lasting changes in pay from temporary fluctuations that may have disappeared by the SCE's twelve-month horizon.

#### 2.3.1 The Earnings Benchmark

The benchmark process for adjusted log hourly earnings, $z_{it}$, is

$$z_{it} = p_{it} + \theta_{it}, \qquad p_{it} = p_{i,t-1} + \psi_{it}. \tag{1}$$

The permanent shock $\psi_{it}$ accumulates through the random walk $p_{it}$ and captures lasting changes in pay, such as the unanticipated part of a merit raise or cost-of-living adjustment. For a worker receiving such a raise, the increase carries through to her earnings in subsequent months. By contrast, a one-off discretionary bonus raises her earnings in the month of receipt without changing her regular pay rate. The transitory component $\theta_{it}$ captures these temporary fluctuations. Both shocks have mean zero, are i.i.d. across months and independent of each other, and have constant variances within each income group.

The permanent and transitory variances can be recovered from how earnings changes follow one another. A temporary payment raises earnings when it arrives and lowers them when it disappears, creating a negative relationship between consecutive changes. A permanent pay rise also raises earnings on receipt, but does not produce the same reversal the following month. The decomposition uses this difference to distinguish temporary fluctuations from lasting changes in pay. In particular, consider the monthly change $d_{it} = \psi_{it} + \theta_{it} - \theta_{i,t-1}$ and let $S_g = \mathbb{E}[d_{it}^2 \mid g]$ denote the second moment of monthly changes in income group $g$. Under the benchmark assumptions,

$$\mathcal{T}_g = -\mathbb{E}[d_{it} d_{i,t-1} \mid g], \qquad \mathcal{P}_g = S_g - 2\mathcal{T}_g, \tag{2}$$

where $\mathcal{T}_g$ is the temporary variance and $\mathcal{P}_g$ the permanent-innovation variance. I estimate these moments using survey-weighted averages of monthly changes and consecutive change pairs within the same job, allowing reported hours to change.

The SCE asks about earnings twelve months ahead, conditional on the same job, employer and hours. To compare this with the SIPP, I remove the predictable component of log hourly earnings and study changes in the residual, $z_{it}$.[4] This focuses the comparison on earnings movements around a predictable wage path. Under the benchmark process, the twelve-month change variance

[4] I obtain $z_{it}$ as the residual from a Mincerian regression of log hourly earnings on age, age squared, sex, race and education, with state and calendar year-month effects. For the comparison with the SCE, I treat the fitted earnings change over the forecast horizon as known to the worker.

is

$$\mathrm{Var}(z_{i,t+12} - z_{it} \mid g) = 12\mathcal{P}_g + 2\mathcal{T}_g. \tag{3}$$

Permanent shocks accumulate over the twelve months, while temporary pay enters through earnings in the initial and final months. The SIPP also shows dependence between earnings changes beyond one month, which the simple process does not capture (Appendix Table C.2). The twelve-month variance is therefore a model-based benchmark; Appendix Table C.3 also examines earnings changes at annual reporting boundaries.

### 2.3.2 Differences across Income Groups

The twelve-month earnings-change benchmark is approximately flat across income groups, while perceived risk decreases with income. Permanent-innovation variation is 8.93 in the lower-income group and 8.39 in the higher-income group, while temporary variation rises from 2.05 to 4.06. The higher–lower difference is therefore statistically significant for temporary variation ($p < 0.001$), but not for the implied twelve-month variance ($p = 0.467$). Guvenen et al. (2021) also find rising annual earnings-growth dispersion at the top among job-stayers, attributable to temporary variation. Higher-income workers thus report less uncertainty than lower-income workers ($p < 0.001$), despite experiencing similar twelve-month earnings variation in the SIPP comparison.

**Table 2:** Perceived risk and realised earnings variation by income group

| | SIPP | | | | SCE |
|---|---|---|---|---|---|
| Income group | Monthly variation, $S_g$ | Permanent $\mathcal{P}_g$ | Transitory $\mathcal{T}_g$ | Implied 12-month change variance | Perceived variance |
| Lower | 13.02 (0.26) | 8.93 (0.24) | 2.05 (0.11) | 111.21 (2.76) | 20.62 (1.33) |
| Middle | 14.75 (0.27) | 8.64 (0.18) | 3.06 (0.13) | 109.75 (2.15) | 11.33 (0.64) |
| Higher | 16.51 (0.27) | 8.39 (0.17) | 4.06 (0.13) | 108.76 (1.96) | 11.41 (0.37) |
| Wald $p$-value | $< 0.001$ | 0.061 | $< 0.001$ | 0.467 | $< 0.001$ |

*Notes:* Survey-weighted estimates; hours may change within SIPP job spells. SCE variance is in squared percentage points; SIPP log variances are scaled by $10^4$. Standard errors in parentheses cluster by respondent (SCE) or household (SIPP). Wald tests compare higher and lower incomes. See Appendix D.3.

At the twelve-month horizon, perceived variance amounts to 19 percent of the implied earnings-change variance in the lower-income group, compared with about 10 percent in both the middle- and higher-income groups. The proportional gap therefore widens with income. Wang (2026) similarly finds that wage-risk perceptions in the SCE fall well below conventional estimates from earnings panels. I will therefore focus on relative differences in these ratios across income groups to mitigate reporting error in the SIPP benchmark, under the assumption that it inflates each group's benchmark by the same proportion. In addition, how much of this gap reflects mistaken beliefs remains an open question. Workers can anticipate payments that the earnings model treats as shocks (Pistaferri, 2001), while measurement error can distort recorded earnings. For instance, Bound and Krueger (1991) document serially correlated errors in earnings reports, so reporting error may affect both components of the decomposition.

### 2.3.3 Persistence

Perceived risk is also more persistent within respondents at lower incomes. In particular, I examine whether a worker who reports more uncertainty than usual this month continues to do so in subsequent interviews. I estimate a first-order autoregression in adjusted perceived variance for each income group, with fixed effects for each uninterrupted response spell. Because the SCE follows respondents for only a short period, the usual fixed-effects estimator is subject to Nickell bias. I therefore use the system GMM estimator of Blundell and Bond (1998), which uses earlier reports as instruments for lagged perceived variance. Ma et al. (2020) similarly use dynamic-panel GMM to address short-panel bias when estimating persistence in managerial forecast errors. Table 3 reports a monthly persistence coefficient of 0.343 for the lower-income group, compared with 0.142 and 0.144 for the middle- and higher-income groups. The higher–lower difference is statistically significant ($p < 0.001$).

For the SIPP, I use a standard proxy for earnings volatility.[5] Its persistence measures whether large residual earnings movements, of either sign, tend to cluster over time. Using the same system GMM specification as for perceived risk, I find the opposite income gradient: the coefficient rises from 0.193 in the lower-income group to 0.235 in the higher-income group ($p = 0.007$). The opposing gradients remain under alternative instrument lags, although the Hansen tests reject the moment restrictions for all three SIPP groups and the middle-income SCE group (Appendix Tables D.1–D.2).

**Table 3:** Monthly persistence of perceived risk and the SIPP volatility proxy

| Income group | SIPP volatility proxy | SCE risk reports |
|---|---|---|
| Lower | 0.193 (0.013) | 0.343 (0.046) |
| Middle | 0.187 (0.010) | 0.142 (0.026) |
| Higher | 0.235 (0.009) | 0.144 (0.025) |
| Wald $p$-value | 0.007 | $< 0.001$ |

*Notes:* Monthly AR(1) coefficients from two-step system GMM. Standard errors in parentheses cluster by response spell and month (SCE) or job spell (SIPP). Wald tests compare higher and lower incomes. Appendices D.1–D.2 give the specification and supporting checks.

Taken together, these estimates show that higher-income workers report lower and less persistent perceived risk, despite greater earnings variation and a more persistent volatility proxy in the SIPP. The earnings records do not reveal which payments workers anticipated or recognised as temporary, however. For instance, a worker who recognises an unexpected bonus can distinguish it from a lasting change in regular pay, even though the payment still creates variation in her recorded earnings. The central question of this paper is therefore whether differences in workers' information about the components of their pay can help explain these opposing patterns. This requires understanding which payments generate the observed earnings variation and how workers might interpret and anticipate them.

[5] I estimate $d_{it} = \alpha + \rho d_{i,t-1} + u_{it}$ by pooled, survey-weighted least squares, where $d_{it} = \Delta z_{it}$ is the adjusted monthly earnings change. The volatility proxy $\widehat{u}_{it}^2$ captures the size of movements not explained by the previous month's change, regardless of sign.

# 3 Properties of Pay Components

The same increase in monthly earnings can have different implications for future pay depending on its source. For instance, a worker who recognises an increase as a bonus may expect her regular wage to remain unchanged, even if she expects the bonus to recur. Information about commissions or occasional overtime may help her make a similar distinction. I compare the contributions of the different pay components and show that bonuses account for almost all of the increase in measured transitory earnings variation from lower to higher incomes. I therefore examine the duration and recurrence of these payments, with a focus on bonuses, to understand what workers could anticipate about these payments and what they could learn from them about future pay.

### 3.1 Contributions to Earnings Variation

I start by decomposing the transitory earnings variation reported in Table 2 into contributions from wages and salaries, bonuses, commissions, tips and overtime. I retain the same adjustment, trimming and same-job sample, allowing hours to change. Let $d_{it}^k$ denote component $k$'s contribution to the adjusted log earnings change. I construct these contributions so that they add up to the total change, $d_{it} = z_{it} - z_{i,t-1} = \sum_k d_{it}^k$.[6]

Each component's contribution includes its own earnings movements and its comovements with other payments. I split each cross-category term equally between the two categories. Using the same moment convention as in equation (2), component $k$ receives

$$\begin{aligned}\mathcal{T}_{kg} &= -\frac{1}{2}\mathbb{E}\left[d_{it}^k d_{i,t-1} + d_{it} d_{i,t-1}^k \mid g\right] \\ &= -\underbrace{\mathbb{E}[d_{it}^k d_{i,t-1}^k \mid g]}_{\text{own-category term}} - \frac{1}{2}\sum_{\ell \neq k} \underbrace{\mathbb{E}\left[d_{it}^k d_{i,t-1}^\ell + d_{it}^\ell d_{i,t-1}^k \mid g\right]}_{\text{cross-category terms}}. \end{aligned} \tag{4}$$

The own-category term captures the tendency for a payment increase to be followed by a decline in the same component. The remaining terms account for movements across categories.

[6] I construct the contributions from changes in adjusted payment amounts, taking logarithms only of total pay. This allows months with zero bonuses or other payments to remain in the decomposition (Appendix E.1).

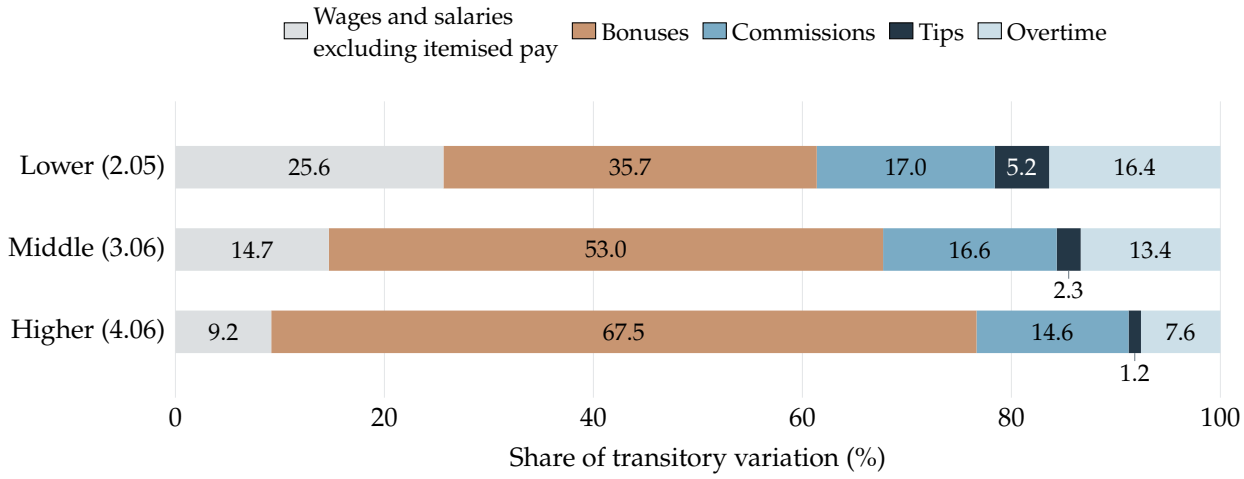



**Figure 2:** Pay components' shares of transitory variation

*Notes:* SIPP, 2013–2023, using the hours-may-change sample of Table 2. Shares include cross-category terms. Group totals appear in parentheses, with log moments scaled by $10^4$.

*Finding 1: Bonuses account for the increase in measured transitory earnings variation with income.* Figure 2 shows that their share rises from 35.7 percent at lower incomes to 67.5 percent at higher incomes, a significant difference ($p < 0.001$). The bonus contribution rises from 0.73 to 2.74, while the combined contribution of the other four components remains close to 1.3 in both groups. If workers recognise bonuses as temporary, the greater bonus share at higher incomes may help them distinguish temporary payments from changes in permanent pay.

To check whether the shift towards bonuses extends to lasting earnings changes, I report the corresponding decomposition of permanent-innovation variation in Appendix Figure E.1. Wages and salaries excluding itemised pay account for at least three-quarters of permanent variation in every income group. The bonus share nevertheless rises from 4.7 percent at lower incomes to 12.4 percent at higher incomes. The bonus contribution therefore rises with income in both components, but the absolute increase is much larger for transitory variation.

### 3.2 Payment Duration

I next examine how the different pay components evolve after receipt, to understand what workers could learn from the source of an earnings increase. For instance, a worker who recognises a spike in pay as a bonus may expect it to disappear without revising her view of regular pay. When additional payments continue for several months, distinguishing them from a lasting pay rise using total earnings alone may be harder.

To follow payments from a new start, I focus on months preceded by at least two consecutive months without the relevant component. This helps distinguish a new overtime spell, for instance, from one already under way. The two-month rule excludes restarts after a single missed month while still admitting quarterly payments. With either one or three prior zero months, bonuses still fall away quickly while overtime remains positive through month six (Appendix Table F.1).

I estimate local projections to examine how each component's current payment relates to its payments over the next six months. For component $k$, let $x^k_{i,t+h}$ denote its hourly amount in

month $t + h$, scaled by the worker's average total hourly earnings over the twelve months before $t$. Holding this scale fixed across horizons, I estimate

$$x^k_{i,t+h} = \beta^k_h x^k_{it} + C'_{it}\gamma^k_h + \tau^k_{t,h} + \alpha^k_{g,h} + \varepsilon^k_{i,t+h}, \qquad h = 0, \ldots, 6. \tag{5}$$

where $C_{it}$ contains twelve months of earnings and payment history, log average prior hourly earnings and log hours in the preceding month. The calendar-date effects $\tau^k_{t,h}$ account for shared payment timing, such as year-end bonuses, while the income-group effects $\alpha^k_{g,h}$ allow average payments to differ across groups. After accounting for these patterns and the worker's prior payment history, $\beta^k_h$ measures whether a larger payment today is followed by larger payments $h$ months later.

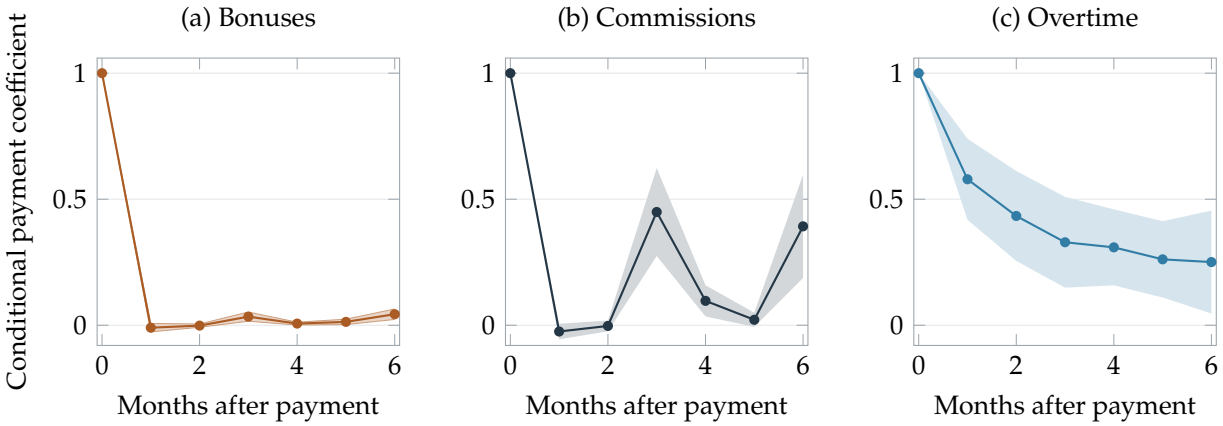



**Figure 3:** Payment profiles following two months without receipt

*Notes:* SIPP, 2013–2023. I pool income groups because there are too few new payment episodes to estimate each component's path precisely within every group. Workers remain in the same recorded job throughout the observation window, while their hours may change. Coefficients relate future to current normalised hourly payments; the initial coefficient is one by construction. Shading gives 95% pointwise confidence intervals with standard errors clustered by worker.

*Finding 2: Bonuses fall away quickly, commissions show quarterly recurrence, and overtime payments persist for several months.* Figure 3 shows that the bonus coefficient is close to zero in the following month, with small positive coefficients at months three and six, suggesting that bonuses mainly provide one-off additions to monthly pay. Commissions instead display pronounced peaks of 0.45 and 0.39 at these horizons, consistent with quarterly payments: the additional pay can disappear between payment dates and return at regular intervals. For overtime, the coefficient remains 0.58 after one month and 0.25 after six months. This gradual decline is consistent with spells of extra work that sustain additional earnings for several months. Too few workers begin receiving tips after a payment gap to estimate their duration reliably. I extend the profiles to twelve months in Appendix Figure F.1, which shows a pronounced annual bonus peak (0.433, SE 0.041). The estimates for commissions and overtime at this horizon are less precise, with overtime only just significant at the 5% level.

These payment profiles suggest that knowing the source of an earnings increase may help workers assess whether their regular pay has changed. For instance, a worker who recognises a bonus as temporary can set it aside when assessing her regular wage, even if she did not anticipate

receiving it. Overtime can instead sustain earnings for several months without a permanent pay rise, making the distinction harder to infer from total pay alone. Since bonuses account for more temporary earnings variation at higher incomes, recognising their temporary nature may be particularly valuable for these workers.

### 3.3 Annual Bonus Recurrence

Bonus history may also help workers anticipate future payments. I examine whether last year's bonus helps predict both the receipt and amount of a payment in the same month this year. For instance, a worker who regularly receives a December bonus may anticipate the rise in pay that month and the fall afterwards, while remaining uncertain about whether the payment will arrive and how large it will be. These anticipated earnings movements still contribute to measured variation, even though they need not generate uncertainty for the worker.

For bonus receipt, I estimate the following regression separately within each income group $g$, with $B_{it}$ indicating whether a bonus was received:

$$B_{it} = c_{i,j(i,t)} + \tau_t + \gamma_g B_{i,t-12} + \eta_g B_{i,t-1} + \zeta_g F_{it} + \varepsilon_{it}, \tag{6}$$

where $F_{it} = \frac{1}{11}\sum_{s=1}^{11} B_{i,t-s}$ controls for bonus receipt frequency over the preceding eleven months, and $B_{i,t-1}$ controls for receipt last month. Together, these help distinguish annual recurrence from bonuses paid frequently or in consecutive months. The worker–job effects $c_{i,j(i,t)}$ account for some jobs routinely paying bonuses and others rarely doing so, while the calendar-date effects $\tau_t$ account for common payment timing, such as bonuses being more frequent in December. The annual coefficient $\gamma_g$ then measures whether a worker who received a bonus last December is more likely to receive one this December, beyond the general increase in bonus receipt that month and conditional on her job and recent payment history.

I repeat the regression using monthly bonus amounts, replacing the receipt indicators with amounts and $F_{it}$ with their preceding eleven-month average.

**Table 4:** Annual recurrence in bonus receipt and amounts

| Association with last year's bonus | Lower | Middle | Higher | Higher–lower $p$-value |
|---|---|---|---|---|
| Receipt probability (pp) per receipt last year | 6.9 (3.4) | 7.5 (1.8) | 20.6 (1.3) | $< 0.001$ |
| Amount (\$) per \$100 more last year | 2.75 (2.29) | 7.34 (5.61) | 29.83 (3.22) | $< 0.001$ |

*Notes:* Survey-weighted regressions with worker-clustered standard errors in parentheses. Wald tests compare higher- and lower-income annual coefficients. Amounts are in nominal dollars. Both exercises require a complete preceding year in the same recorded job and allow hours to change.

*Finding 3: Bonus receipt shows stronger annual recurrence at higher incomes.* Table 4 shows that receipt in the same month last year is associated with a 6.9 percentage-point increase in current receipt probability at lower incomes, compared with 20.6 points at higher incomes ($p < 0.001$ for the

difference). Commissions, tips and overtime do not show the same positive annual association in receipt after controlling for recent payment history (see Appendix Table G.1, panel A). For bonuses, the higher-income advantage is specific to the twelve-month lag among the horizons examined. At three months, recurrence is slightly stronger at lower incomes, while at six months the higher–lower difference is not statistically significant (see Appendix Table G.2). The pattern therefore points to annual timing as a distinctive feature of bonus receipt at higher incomes.

The baseline amount regressions show a similar annual gradient: an additional $100 in last year's same-month bonus is associated with $2.75 more today at lower incomes and $29.83 more at higher incomes. The amount regressions for commissions, tips and overtime likewise show no positive annual association after controlling for recent payment history (see Appendix Table G.1, panel B). One possible explanation, consistent with the payment profiles in Figure 3, is that overtime follows spells of extra work lasting several months without recurring annually, while commissions follow shorter, quarterly payment cycles, but ultimately don't contribute to the annual result.

**Table 5:** Bonus timing and earnings variation

| | Lower | Middle | Higher | Higher–lower $p$-value |
|---|---|---|---|---|
| Months with bonus receipt one year earlier (%) | 0.58 (0.07) | 1.14 (0.07) | 1.62 (0.07) | $< 0.001$ |
| Share of bonus variation from these months (%) | 17.9 (6.4) | 33.7 (4.3) | 32.5 (2.3) | 0.032 |
| Share of total temporary variation from bonuses in these months (%) | 6.6 (2.4) | 19.4 (3.0) | 23.5 (1.9) | $< 0.001$ |

*Notes:* Survey-weighted estimates allowing hours to change, with the required same-job payment history. Worker-clustered analytical standard errors are in parentheses. Wald tests compare higher and lower incomes. Appendix G.4 describes the calculation.

Bonus history also identifies months containing a substantial share of earnings variation. At higher incomes, months with bonus receipt in the same job one year earlier represent only 1.6% of eligible observations, yet their bonus contribution accounts for 23.5% of total temporary variation (Table 5). At lower incomes, the corresponding shares are 0.6% and 6.6%.

These results motivate a separate bonus component that can recur in the same calendar month each year. For instance, a worker who receives a bonus this month may expect another twelve months ahead, at the SCE's forecast horizon. Her uncertainty about future earnings then depends on what her bonus history tells her about the receipt and size of that payment, as well as uncertainty about the rest of her pay.

### 3.4 Forecasting Earnings with Bonus Information

Lastly, I test whether knowing how much of a worker's pay comes from bonuses improves forecasts of her future earnings beyond the information in total pay. For instance, a worker who receives an unusually high payment may infer a smaller rise in regular pay if she knows that part of it is a bonus. I compare forecasts based on current and past total earnings with forecasts that also use

the bonus share of those payments.

For log hourly earnings $w_{it}$, I estimate the following regression separately by income group $g$ and forecast horizon $h$:

$$w_{i,t+h} - w_{it} = \alpha_{gh} + X'_{it}\gamma_{gh} + \boldsymbol{c}'_{it}\pi_{gh} + u_{it,h}, \qquad h \in \{3, 6, 12\}. \tag{7}$$

The baseline regression omits $\boldsymbol{c}_{it}$ and uses only $X_{it}$: current log hourly earnings, their three monthly lags, their average over earlier observations in the same job, and calendar-month and year controls. The additional variables $\boldsymbol{c}_{it}$ contain the current bonus share, its three monthly lags and its average over earlier observations in the same job. Each share divides bonus pay by total pay in that month. The comparison therefore asks whether the payment breakdown adds information beyond the level and recent path of total earnings.

Since bonus variables can improve fit simply by capturing noise, I split workers into five sets and predict each using regressions estimated on the other four. Both specifications use the same observations and survey weights, and I measure improvement as the percentage reduction in mean squared forecast error.

**Table 6:** Forecast gains from bonus information

| Forecast horizon | Lower | Middle | Higher | Higher–lower $p$-value |
|---|---|---|---|---|
| 3 months | 0.40 (0.12) | 0.96 (0.12) | 3.82 (0.18) | $< 0.001$ |
| 6 months | 0.14 (0.06) | 0.38 (0.06) | 1.81 (0.11) | $< 0.001$ |
| 12 months | −0.01 (0.01) | −0.04 (0.04) | 0.21 (0.06) | $< 0.001$ |

*Notes:* Percentage reductions in survey-weighted mean squared forecast error. Approximate worker-clustered standard errors in parentheses condition on fitted forecasts; Wald tests compare higher and lower incomes. Workers remain in the same recorded job; hours may change. Samples vary by horizon. Appendix H.1 gives further details.

*Finding 4: Bonus information improves earnings forecasts more at higher incomes, particularly over the next three to six months.* Table 6 shows that adding bonus information reduces three-month mean squared forecast error by 3.82 percent at higher incomes, compared with 0.40 percent at lower incomes ($p < 0.001$). The gain declines with the forecast horizon but remains significant. At twelve months, it amounts to only 0.21 percent at higher incomes, with no significant improvement in the other two groups.

To separate the contributions of current and past bonus information, I also estimate forecasts that add only past bonus shares to the total-pay baseline. These produce little improvement over three and six months; most of the gain appears when I also include the current bonus share (Appendix Table H.1). When I add the other pay components at the twelve-month horizon, the forecast gain at higher incomes rises from 0.21 to 0.43 percent ($p = 0.025$ for the additional gain), suggesting that information about future earnings extends beyond bonuses, although the annual gains remain modest (Appendix Table H.2).

The near-term gains are consistent with current bonus information helping workers distinguish

temporary payments from changes in regular pay. How far this can explain the perceived-risk gradient depends on how workers use that information, which the forecast comparison alone does not establish.

# 4 Pay Information and Earnings Risk

I have shown that bonus information improves earnings forecasts, particularly at higher incomes. To examine how information about temporary payments affects workers' earnings uncertainty, I develop a simple Bayesian learning model. The worker observes total earnings and may separately identify part of her temporary pay and know when some of these payments recur. Bonuses motivate this information structure, which can also describe other temporary payments under the same assumptions. The Kalman filter provides a tractable starting point, explaining how workers infer permanent pay and how uncertainty about its level contributes to perceived earnings risk. I introduce this learning problem in a Gaussian setting and derive propositions connecting the empirical findings on bonuses to workers' beliefs about future earnings. A quantitative model then examines whether these effects can explain the income gradients in the level and persistence of perceived earnings risk.

## 4.1 Environment

### 4.1.1 Earnings Process

I retain the permanent–transitory structure of the earnings benchmark and, crucially, distinguish temporary pay that the worker can separately identify, such as bonuses, from the remaining temporary pay:

$$w_t = p_t + u_t + b_t, \qquad p_t = p_{t-1} + \psi_t, \tag{8}$$

where $w_t$ denotes adjusted log hourly earnings in month $t$, $p_t$ is permanent pay, $b_t$ is a temporary component that the worker can identify separately and $u_t$ captures the remaining temporary pay. Bonuses provide an example when the worker can observe the amount and recognise it as temporary. The shock $\psi_t$ permanently changes pay, while $u_t$ and $b_t$ affect earnings only in the month received. In the baseline, the temporary components are centred Gaussian terms, $u_t \sim N(0, \sigma_u^2)$ and $b_t \sim N(0, \sigma_b^2)$, with constant variances. Their one-month duration approximates the short-lived bonus payments in Figure 3.

For parsimony, I abstract from recurrence in other payments and the longer duration of overtime, since their combined contribution to transitory variation changes little across income groups. Allowing for these dynamics would be an interesting extension and left for future work. The baseline also leaves aside annual bonus recurrence. I outline an optional extension below and examine it in the quantitative model in Section 5.

I allow permanent-pay risk to vary over time through a monthly version of the ARCH specification in Meghir and Pistaferri (2004), where $q_t$ denotes the variance of the permanent shock:

$$\psi_t = \sqrt{q_t}\,\varepsilon_t, \qquad q_{t+1} = (1-\alpha)\bar{q} + \alpha\psi_t^2, \qquad \varepsilon_t \sim N(0,1), \tag{9}$$

where $\bar{q} > 0$ is the long-run mean of $q_t$ and $0 \leq \alpha < 1$. For $\alpha > 0$, a worker who experiences a

large permanent pay rise or cut faces greater risk of another large change, although its direction remains uncertain. A higher $\alpha$ makes this elevated risk more persistent; at $\alpha = 0$, risk is constant at $\bar{q}$. The shocks $\varepsilon_t$, $u_t$ and $b_t$ are independent across components and months, and independent of the initial states, conditional on the parameters.

Motivated by the annual bonus peak in Appendix Figure F.1 and the recurrence estimates in Section 3.3, I also allow bonus payments to recur annually. The worker may then know when a temporary payment will arrive, while remaining uncertain about its amount. The same idea can describe what workers know about other scheduled payments, such as government transfers or holiday pay. To model this recurrence, I split bonus pay into irregular monthly bonus variation and an annual payment, writing $b_t = v_t + D_t a_t$, with $v_t \sim N(0, \sigma_v^2)$ and $a_t \sim N(0, \sigma_a^2)$. I let a share $\rho$ of workers receive an annual payment. For these workers, $D_t = 1$ in their annual payment month and zero otherwise. For the remaining workers, it is always zero. I retain the Gaussian and independence assumptions of the baseline for $\varepsilon_t$, $u_t$, $v_t$ and $a_t$, conditional on the parameters and payment schedule.

### 4.1.2 Worker Information and Perceived Risk

Building on the income-learning framework of Druedahl and Jørgensen (2020), I assume that the worker knows the form of this process but observes neither permanent pay $p_t$ nor its risk $q_t$. She must infer both from her payments and learn the fixed parameters $\vartheta = (\bar{q}, \alpha, \sigma_u^2, \sigma_b^2)$ in the baseline. With recurring payments, she instead learns the variances of irregular and annual bonuses, $\sigma_v^2$ and $\sigma_a^2$, together with the payment schedule when it is not observed. For simplicity, all workers start with a common prior over $\vartheta$, so income differences in beliefs are not imposed at the outset.[7] The true parameters differ across groups, and workers learn about them from their own payments using Bayes' rule. Their posteriors become more precise on average as evidence accumulates.

To isolate the role of information about temporary pay within each income group, I compare beliefs formed from the same earnings history under three nested information sets with common initial information $\mathcal{I}_0$. Observing total pay gives the worker $\mathcal{I}_t^0 = \mathcal{I}_0 \cup \{w_s\}_{s=1}^t$, under which she uses the signal $y_t^0 = w_t$. Observing bonus amounts expands her information to $\mathcal{I}_t^1 = \mathcal{I}_t^0 \cup \{b_s\}_{s=1}^t$, allowing her to recognise and subtract the temporary bonus, leaving $y_t^1 = w_t - b_t = p_t + u_t$. She still has to distinguish permanent pay from the remaining temporary component $u_t$, but faces less noise in doing so. Knowing her payment schedule further expands her information to $\mathcal{I}_t^2 = \mathcal{I}_t^1 \cup \{C\}$, where $C = 0$ denotes no annual payment and $C \in \{1, \ldots, 12\}$ its payment month. She must still learn how much irregular and annual bonuses fluctuate.

The main objects of interest are therefore the differences in the level and persistence of perceived earnings risk between these information sets. I define this risk as

$$V_{t,12}^j = \mathrm{Var}(w_{t+12} \mid \mathcal{I}_t^j) = P_t^j + \mathbb{E}[\sigma_u^2 + \sigma_v^2 + D_{t+12}\sigma_a^2 \mid \mathcal{I}_t^j] + \sum_{h=1}^{12} \mathbb{E}[q_{t+h} \mid \mathcal{I}_t^j], \tag{10}$$

to match the SCE's twelve-month forecast horizon. Here, $P_t^j = \mathrm{Var}(p_t \mid \mathcal{I}_t^j)$ measures her uncer-

[7] Conditional on the parameters, initial permanent pay has a Gaussian prior and initial risk is drawn from the long-run distribution implied by the ARCH process. The prior gives initial pay a finite second moment and the variance parameters finite means, ensuring that earnings forecast variances are well defined.

tainty about current permanent pay. Future permanent shocks accumulate over the year, while temporary payments contribute uncertainty only at the forecast date. In the baseline without recurring payments, the temporary-pay term is $\mathbb{E}[\sigma_u^2 + \sigma_b^2 \mid \mathcal{I}_t^j]$.

### 4.2 Bayesian Learning

The worker must infer current permanent pay and learn how much permanent and temporary pay fluctuate. These inferences are linked: expected risk affects how a worker interprets a pay surprise, while the permanent change she infers provides evidence about future risk. I first explain inference about permanent pay with known, constant variances, then turn to learning about earnings risk and scheduled payments.

#### 4.2.1 Learning about Permanent Pay

To explain the Kalman update in the baseline without recurring payments, I temporarily assume that the worker perfectly knows the variance parameters and set $\alpha = 0$, so $q_t = \bar{q}$. This means that she knows how much permanent and temporary pay fluctuate, but still has to infer how much of each payment is permanent. I relax the known-variance assumption when turning to learning about earnings risk.

I follow the standard Kalman filtering procedure. Before the payment at date $t$, she expects permanent pay to equal her previous estimate, $\widehat{p}_{t-1}^{\,j} = \mathbb{E}[p_{t-1} \mid \mathcal{I}_{t-1}^j]$, since the new permanent shock has mean zero. The shock nevertheless adds uncertainty, so the prediction variance is $P_{t|t-1}^j = P_{t-1}^j + \bar{q}$, where $P_{t-1}^j = \mathrm{Var}(p_{t-1} \mid \mathcal{I}_{t-1}^j)$ measures her remaining uncertainty about last month's permanent pay.

The payment then provides a noisy signal of permanent pay. With total-pay information, the temporary noise has variance $R_0 = \sigma_u^2 + \sigma_b^2$; recognising the bonus reduces it to $R_1 = \sigma_u^2$. On observing $y_t^j$, she combines this signal with her previous estimate using the Kalman update:

$$\widehat{p}_t^{\,j} = \widehat{p}_{t-1}^{\,j} + K_t^j \underbrace{(y_t^j - \widehat{p}_{t-1}^{\,j})}_{\text{pay surprise}}, \qquad K_t^j = \frac{P_{t|t-1}^j}{P_{t|t-1}^j + R_j}, \qquad P_t^j = \underbrace{(1 - K_t^j)}_{\text{share retained}} P_{t|t-1}^j. \tag{11}$$

Here $\widehat{p}_t^{\,j}$ is her revised estimate of permanent pay, $P_t^j$ measures her remaining uncertainty about it, and $e_t^j = y_t^j - \widehat{p}_{t-1}^{\,j}$ denotes the pay surprise. The Kalman gain $K_t^j$ is the weight she places on the pay surprise: she relies more on the payment when her previous estimate is uncertain, and less when temporary noise makes the payment harder to interpret. For a given prediction variance, recognising bonuses lowers $R_j$, raises $K_t^j$ and hence reduces $P_t^j$.

Bonus information also changes the pay surprise itself. For instance, a worker who can attribute an unusually high payment to a bonus has little reason to revise her estimate of permanent pay upwards. Overall, recognising the bonus can make the worker more certain about permanent pay without requiring a large revision to her estimate of its level.

### 4.2.2 Learning about Earnings Risk

I now allow the worker to learn the variance parameters and return to the time-varying risk process in equation (9). Since a large permanent pay rise or cut raises the risk of another large change when $\alpha > 0$, the worker's interpretation of a pay surprise also informs her assessment of future shock variances, affecting both the level and persistence of perceived earnings risk.

After observing each payment, she uses Bayes' rule to update her beliefs about the permanent shock and the parameters governing the earnings process. Since she knows the ARCH structure, she obtains her estimate of next month's permanent-shock variance, $\widehat{q}^{\,j}_{t+1|t} = \mathbb{E}[q_{t+1} \mid \mathcal{I}_t^j]$, by averaging the ARCH equation over her updated beliefs:

$$\widehat{q}^{\,j}_{t+1|t} = \mathbb{E}\Big[(1-\alpha)\bar{q} + \alpha\psi_t^2 \mid \mathcal{I}_t^j\Big]. \tag{12}$$

Because the permanent shock is unobserved, she must account for both her estimate of its size and her uncertainty about it. The identity $\mathbb{E}[\psi_t^2 \mid \mathcal{I}_t^j] = (\mathbb{E}[\psi_t \mid \mathcal{I}_t^j])^2 + \mathrm{Var}(\psi_t \mid \mathcal{I}_t^j)$ shows why using only the squared estimate would omit relevant uncertainty. For instance, a worker whose estimated permanent shock is close to zero may still consider sizeable positive and negative shocks plausible. Her expected squared shock can therefore remain large even when her estimated change in permanent pay is small.

Bonus information therefore changes what the worker learns about earnings risk. Observed bonus amounts inform her beliefs about $\sigma_b^2$, while subtracting them helps her assess permanent shocks from the remaining pay. For instance, recognising an unusually large payment as a bonus can weaken the evidence that permanent pay has become riskier. Over successive payments, this distinction also affects what she learns about the long-run permanent variance $\bar{q}$ and the persistence of risk $\alpha$.

### 4.2.3 Learning about Scheduled Payments

Under $\mathcal{I}_t{}^2$, the worker knows when her annual payment arrives. Outside that month, she observes only irregular bonus variation and learns about $\sigma_v^2$. In the annual payment month, her bonus includes both components. By accounting for the irregular variation, she also learns about annual-payment variance, $\sigma_a^2$. A large annual payment can therefore raise her assessment of annual-payment variability without implying the same increase in bonus risk throughout the year.

Knowing the date then helps her assess which future payments matter for her forecast. For instance, a worker expecting a December bonus includes uncertainty about next December's payment in her December report. In January, she forecasts the following January, which contains no annual payment. Her bonus contribution to twelve-month perceived risk is therefore

$$B^2_{t,12} = \mathbb{E}[\sigma_v^2 \mid \mathcal{I}_t^2] + D_{t+12}\mathbb{E}[\sigma_a^2 \mid \mathcal{I}_t^2]. \tag{13}$$

The bonus contribution to reported risk therefore falls when the forecast month no longer includes an annual payment, even if the worker's beliefs about bonus variability remain unchanged. I examine how much this contributes to the persistence gradient in the quantitative model, where the worker learns jointly about permanent pay, earnings risk and the parameters.

### 4.3 Implications for Perceived Earnings Risk

For the baseline earnings process in Section 4.1, I derive analytical results showing how information about temporary pay can improve earnings forecasts, lower perceived risk and affect risk updating. I first express the contribution of future permanent shocks in terms of current risk, its long-run level and its persistence.

#### 4.3.1 The Level of Perceived Earnings Risk

**Proposition 1** (Permanent-risk persistence and forecast uncertainty)**.** *Under the baseline earnings process in Section 4.1, twelve-month perceived earnings risk is*

$$\begin{aligned} V^j_{t,12} = P^j_t + \mathbb{E}[\sigma_u^2 + \sigma_b^2 \mid \mathcal{I}^j_t] + 12\mathbb{E}[\bar{q} \mid \mathcal{I}^j_t] \\ + \mathbb{E}\Big[A_{12}(\alpha)(q_{t+1} - \bar{q}) \mid \mathcal{I}^j_t\Big], \end{aligned} \tag{14}$$

*where* $A_{12}(\alpha) = \sum_{h=0}^{11} \alpha^h$ *is the cumulative weight on current risk over the twelve-month horizon. Appendix J.1 derives this decomposition.*

Permanent shocks accumulate over the year, giving the third term, while the final term adjusts for whether current permanent risk is above or below its long-run level. A higher $\alpha$ makes this deviation last longer and therefore gives it more weight in the twelve-month forecast. Since recognising today's bonus helps distinguish temporary pay from permanent shocks, it can affect both uncertainty about current permanent pay and expected future permanent risk. Recognising today's bonus does not reveal future bonus amounts, which still contribute to her earnings uncertainty.

**Proposition 2** (Pay information and perceived earnings risk)**.** *Conditional on the total-pay history, observing the history of* $b_t$ *weakly reduces expected perceived earnings risk under a common earnings process and prior:*

$$V^0_{t,12} - \mathbb{E}[V^1_{t,12} \mid \mathcal{I}^0_t] = \mathrm{Var}(\widehat{p}^1_t \mid \mathcal{I}^0_t) \geq 0. \tag{15}$$

*The reduction is strict whenever the additional information changes the forecast of permanent pay with positive conditional probability. The result follows from the law of total variance, as shown in Appendix J.2.*

The right-hand side measures how much the forecast of permanent pay differs across histories of $b_t$ consistent with the same total-pay history. For instance, learning which payments were bonuses removes this source of uncertainty on average; the result need not hold after every particular bonus observation.[8] The worker's forecast variance is also her expected squared forecast error when she uses her posterior mean to predict earnings. This provides a theoretical counterpart to Finding 4, where bonus information improves earnings forecasts outside the estimation sample. The information comparison also extends to recurring payments and to knowledge of the payment schedule (Appendix J.2).

I next examine how the share of temporary variation the worker can identify affects her earnings uncertainty. To isolate the role of pay composition, I hold permanent and total temporary variance fixed and vary the share the worker can identify.

**Proposition 3** (Pay composition and the level of perceived risk)**.** *Suppose the variance parameters are*

[8] The average reduction in forecast uncertainty follows more generally from nested information sets and does not depend on the information concerning bonuses.

*known, there are no recurring payments and $\alpha = 0$. In the steady state of the Kalman filter,*

$$\begin{aligned} P^j_\infty &= \frac{\sqrt{\bar{q}^2 + 4\bar{q}R_j} - \bar{q}}{2}, \\ V^j_{\infty,12} &= P^j_\infty + 12\bar{q} + \sigma_u^2 + \sigma_b^2, \end{aligned} \tag{16}$$

*where $R_0 = \sigma_u^2 + \sigma_b^2$ and $R_1 = \sigma_u^2$. Holding $\bar{q}$ and positive total temporary variance $R_0$ fixed, a larger separately observed share of temporary variance strictly lowers $V^1_{\infty,12}$ and leaves $V^0_{\infty,12}$ unchanged. Appendix J.3 derives the steady-state variance and establishes the effect of pay composition.*

For instance, two workers can experience equally variable earnings, yet the worker who can identify more of the temporary variation as bonuses faces less noise when estimating permanent pay. Here, future shocks contribute the same risk, so the difference must then come from uncertainty about current permanent pay. This analytical result applies to the filter's steady state without recurring payments and with known, constant variances. I will relax these assumptions in the quantitative model, where workers learn jointly about permanent pay and future risk. As all terms in Proposition 1 can then change, I am able to assess how large the effect is and whether bonus recognition can offset the greater earnings variation at higher incomes.

#### 4.3.2 Risk Updating and Persistence

Recognising temporary payments may also affect how workers revise their assessment of earnings risk. For instance, a worker who cannot identify temporary payments may need more evidence to distinguish a change in permanent-pay risk from temporary noise. Recognising bonuses may therefore help her correct earlier errors in her assessment of risk more quickly.

I use the squared pay surprise, $(e^j_t)^2$, as a noisy reading of risk, since large surprises of either sign can indicate greater variance. To obtain a Kalman update, I approximate the ARCH risk process and the signal constructed from this surprise by a linear Gaussian system. As before, I treat the parameters as known and evaluate the filters in steady state.

Here, $h_t$ approximates permanent-pay risk $q_t$, and $s^j_t$ is the worker's noisy reading of this risk. The approximation preserves the ARCH conditional mean, with persistence $0 < \alpha < 1$ and risk-innovation variance $Q > 0$. The signal has mean-zero noise with some variance $\Omega_j$, which I keep constant in the analytical comparison, and where a smaller $\Omega_j$ allows the worker to assess risk more precisely.[9]

**Proposition 4** (Pay information and the correction of risk assessments)**.** *With known parameters, no recurring payments and steady-state filtering in the Gaussian approximation, the worker updates her assessment of risk according to*

$$\begin{aligned} \widehat{h}^j_{t|t-1} &= (1-\alpha)\bar{q} + \alpha \widehat{h}^j_{t-1}, \\ \widehat{h}^j_t &= \widehat{h}^j_{t|t-1} + K^j_h\big(s^j_t - \widehat{h}^j_{t|t-1}\big), \end{aligned} \tag{17}$$

*where $\widehat{h}^j_t$ is the worker's estimate of current risk and $K^j_h$ is the steady-state Kalman gain. Positive variance in the separately observed component, $\sigma_b^2 > 0$, implies a more precise risk signal, $\Omega_1 < \Omega_0$, and a larger*

[9] I derive $\Omega_j$ from the variance of the squared pay surprise, using the pay filter in Proposition 3 and evaluating risk at its common long-run level $\bar{q}$. The approximation treats state innovations and signal errors as mutually independent, independent across months and independent of the Gaussian initial state (Appendix J.4).

*gain, $K_h^1 > K_h^0$. Holding $\bar{q}$, $\alpha$, $Q$ and total temporary variance fixed, a larger separately observed share of temporary variance raises the gain under $\mathcal{I}_t^1$. Appendix J.4 derives the gain and the error recursion.*

Recognising bonuses removes their variation from the pay signal and reduces uncertainty about previous permanent pay, making new payments more informative about risk. The worker therefore places greater weight on new evidence and corrects earlier errors more quickly. An earlier estimation error carries forward with coefficient $\alpha(1 - K_h^j)$, so a larger gain reduces its influence on subsequent assessments. Whether this also lowers the persistence of reported earnings risk depends on the other components in Proposition 1, which I examine in the quantitative model.

# 5 Quantitative Model

I now examine the quantitative implications of the propositions in Section 4 by relaxing the assumptions of known variance parameters and steady-state filtering, and assessing to what extent payment information can explain income differences in the level and persistence of perceived earnings risk. I externally calibrate the earnings process to the SIPP and use a common learning history across income groups. With information about bonus amounts and payment dates, the model reproduces part of the SCE persistence gap and generates a modest decline in mean perceived risk with income. Given this calibration, I then examine how bonus recognition changes responses to temporary payments and discuss a policy application comparing how annual transfers to households and monthly instalments affect perceived risk.

## 5.1 Quantitative Environment

I simulate monthly earnings histories for workers who remain in the same job. Earnings follow the process in Section 4.1, including the recurring payments:

$$
\begin{aligned}
w_t &= p_t + u_t + b_t, \qquad & p_t &= p_{t-1} + \psi_t, \\
\psi_t &= \sqrt{q_t}\,\varepsilon_t, & q_{t+1} &= (1-\alpha)\bar{q} + \alpha\psi_t^2, \\
b_t &= v_t + D_t a_t. &&
\end{aligned}
$$

The shocks are independent and centred Gaussian, with $\varepsilon_t \sim N(0,1)$ and temporary-payment variances $\sigma_u^2$, $\sigma_v^2$ and $\sigma_a^2$. A share $\rho$ of workers receives an annual payment, with $D_t = 1$ in their scheduled month and zero otherwise. I draw each recipient's payment month uniformly from the twelve calendar months and keep it fixed across years.

I compare each worker's risk assessments under the three information sets, holding her earnings history fixed. She receives the same annual payments in each case, but knows their schedule only under $\mathcal{I}_t^2$. Under $\mathcal{I}_t^0$ and $\mathcal{I}_t^1$, I assume that she does not learn this schedule from past receipts. After each payment, she updates her beliefs jointly about permanent pay, its current risk and the parameters governing her earnings.[10] I then use these beliefs to calculate her twelve-month earnings forecast variance, giving a monthly series of perceived risk under each information set that I can compare with the levels and persistence observed in the SCE.

[10] I explain the numerical implementation of the joint update in Appendix K.1, where I track possible permanent-pay histories and parameter values and revise their probabilities after each payment. Learning about annual-payment variance uses a separate approximation.

To move from the fixed worker types used in the estimation towards a more flexible heterogeneous-agent model, I allow workers to move across the income distribution. I let temporary-pay parameters vary smoothly with permanent income, so the composition of their payments changes with their position in the distribution. For instance, as workers' permanent income rises, a larger share of temporary earnings variation comes from bonuses they can distinguish from regular pay.[11]

### 5.2 Simulation and Calibration

I calibrate the earnings process to the SIPP and specify common initial beliefs across income groups, then simulate workers' risk reports over a common learning history and compare their persistence with the SCE.

#### 5.2.1 Earnings Calibration

I use the SIPP evidence in Sections 2 and 3 to calibrate the earnings process workers face. For each income group, I use the variance decomposition in Section 2.3.1 to determine long-run permanent variance $\bar{q}$ and total temporary variance, then the component decomposition to separate the bonus contribution from other temporary variance $\sigma_u^2$. These targets reflect the approximately flat permanent variance and larger bonus contribution at higher incomes documented in Finding 1. I use the recurrence evidence in Section 3.3 to determine the share of workers receiving annual payments, $\rho$, and split the measured bonus contribution between annual and irregular payments. To calibrate ARCH persistence $\alpha$, I match persistence in the SIPP volatility proxy to the same statistic computed from simulated earnings, obtaining $\alpha = 0.44$.[12] Table 7 reports the earnings parameters, prior assumptions and learning history.

[11] The simulated temporary variances change with permanent income. The worker estimates constant variance parameters and revises her beliefs after each payment, which approximates learning about this changing process. Long-run permanent variance and annual-payment eligibility remain fixed at job entry (Appendix K.6).

[12] The persistence of the SIPP volatility proxy need not equal $\alpha$, since temporary payments also contribute to the proxy. I therefore simulate earnings at different values of $\alpha$ and compare the resulting persistence estimates with the SIPP using the same proxy and estimator (Appendix K.2).

**Table 7:** Calibration and assumptions

| Parameter | Lower | Middle | Higher | Basis |
|---|---|---|---|---|
| *A. Earnings process* | | | | |
| Long-run permanent variance, $\bar{q}$ | 8.93 | 8.64 | 8.39 | SIPP moments |
| Other temporary variance, $\sigma_u^2$ | 1.32 | 1.44 | 1.32 | SIPP moments |
| Average bonus variance, $\sigma_b^2$ | 0.73 | 1.62 | 2.74 | SIPP moments |
| ARCH persistence, $\alpha$ | | 0.44 | | SIPP proxy |
| *B. Annual bonus timing* | | | | |
| Share with annual payments, $\rho$ | 0.070 | 0.136 | 0.194 | SIPP timing |
| Annual-payment variance, $\sigma_a^2$ | 21.9 | 47.0 | 53.1 | SIPP timing |
| Irregular-bonus variance, $\sigma_v^2$ | 0.60 | 1.09 | 1.88 | SIPP timing |
| *C. Common prior beliefs* | | | | |
| Mean permanent variance, $\widetilde{\bar{q}}$ | | 8.65 | | SIPP average |
| Mean monthly temporary variance, $\widetilde{\sigma}_T^2$ | | 3.06 | | SIPP average |
| Log standard deviations, $\tau_q, \tau_T$ | | 2.5, 2.5 | | Assumed |
| Bonus share of monthly temporary variance | | $U(0,1)$ | | Assumed |
| ARCH persistence, $\widetilde{\alpha}$ | | $U(0, 0.55)$ | | Assumed |
| Mean annual-payment variance, $\widetilde{\sigma}_a^2$ | | 40.7 | | SIPP average |
| Annual-variance log standard deviation | | 1.5 | | Assumed |
| *D. Common learning history* | | | | |
| Learning age, $A$ | | 48 | | Benchmark |

*Notes:* Variances are in squared 100-log-point units. Income-specific entries are calibration targets for group averages at job entry.

### 5.2.2 Initial Beliefs

Workers begin with common priors over the variance parameters, so income differences in their beliefs emerge from their own payments rather than income-specific starting values. I set the prior means of long-run permanent and total monthly temporary variance to their SIPP cross-group averages, 8.65 and 3.06. I assume lognormal priors with a common log standard deviation of 2.5, keeping the variances positive and leaving workers with substantial initial uncertainty about earnings risk.

The prior links beliefs about bonus and other temporary variance so that workers can learn about overall temporary-pay variability from the bonuses they observe. Under $\mathcal{I}^0$ and $\mathcal{I}^1$, an uncertain share divides total temporary variance between the two components. A large observed bonus can therefore suggest greater temporary variation overall, leading the worker to revise her assessment of other temporary variance as well. Under $\mathcal{I}^2$, I apply this construction to $\sigma_u^2 + \sigma_v^2$ and specify a separate prior for annual-payment variance $\sigma_a^2$. Workers also learn ARCH persistence $\alpha$, starting from $U(0, 0.55)$.[13]

[13] I set the upper bound at 0.55 to keep the long-run variance of current risk finite. Under the Gaussian ARCH process, this requires $\alpha < 1/\sqrt{3} \approx 0.577$. Under an earlier calibration, I also examine how the results change with the dispersion

### 5.3 Model Fit

I assess the model's fit to persistence across income groups and to their mean perceived risk ratios. The simulation follows 8,192 workers per income group from job entry as they learn from successive payments. To guide the choice of a common learning history $A$, I compare persistence under $\mathcal{I}^2$ with the SCE estimates (Figure 4), using the same system GMM specification on twelve monthly risk assessments starting at payment $A$.

I use persistence to guide the learning history because it responds more strongly to accumulated payment information than mean-risk ratios, which change little across the histories considered. These ratios remain untargeted, providing a separate test of whether measured pay composition and workers' information can explain the SCE income gradient. Appendix Table K.2 reports the full benchmark results under all three information sets.

#### 5.3.1 Persistence of Perceived Earnings Risk

Figure 4 shows that total-pay information, $\mathcal{I}^0$ in panel (a), does not generate systematically lower persistence at higher incomes. The reversal becomes clear in panel (b) when workers know payment dates: persistence under $\mathcal{I}^2$, shown by the solid lines, declines with income. Observing bonus amounts alone, $\mathcal{I}^1$, lowers persistence but does not reproduce this ordering. The distinction therefore concerns what workers know about future payments, beyond recognising the bonus they have just received.

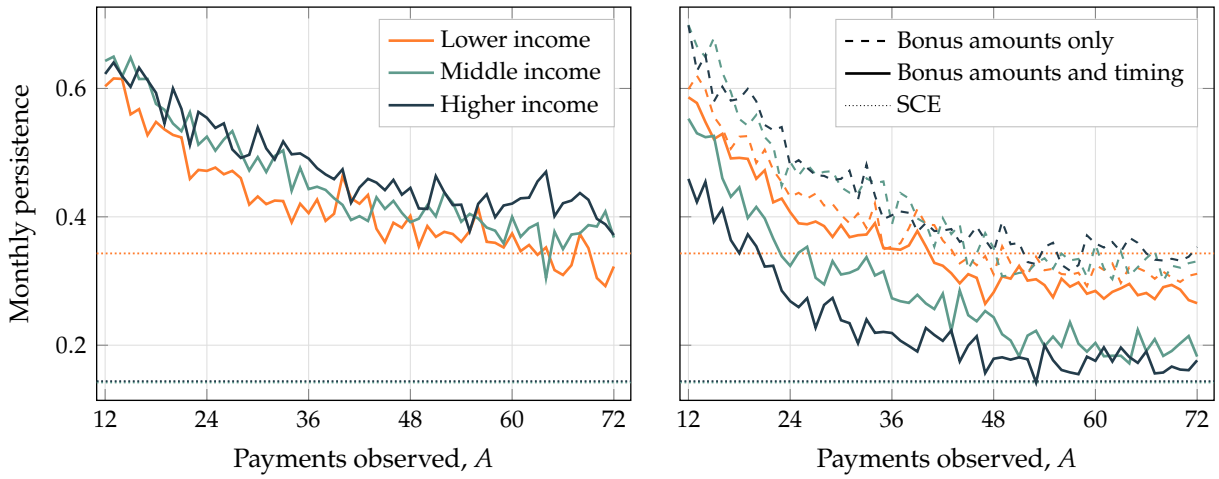



**Figure 4:** Persistence of perceived earnings risk by learning history

*Notes:* Each point is an estimate of monthly perceived-risk persistence from assessments after payments $A$ to $A+11$; adjacent points share eleven assessments. Income groups are reassigned at each $A$ using earnings over payments $A-11$ to $A$. Dotted lines show the SCE estimates.

The decline in persistence with income reflects how workers use information about the recurring bonuses conjectured in Finding 3. To understand this difference, consider a worker receiving a scheduled annual bonus, as illustrated in Appendix Figure K.1. Under $\mathcal{I}^1$, she recognises the receipt as temporary, but a large bonus can lead her to expect greater variation in future bonus

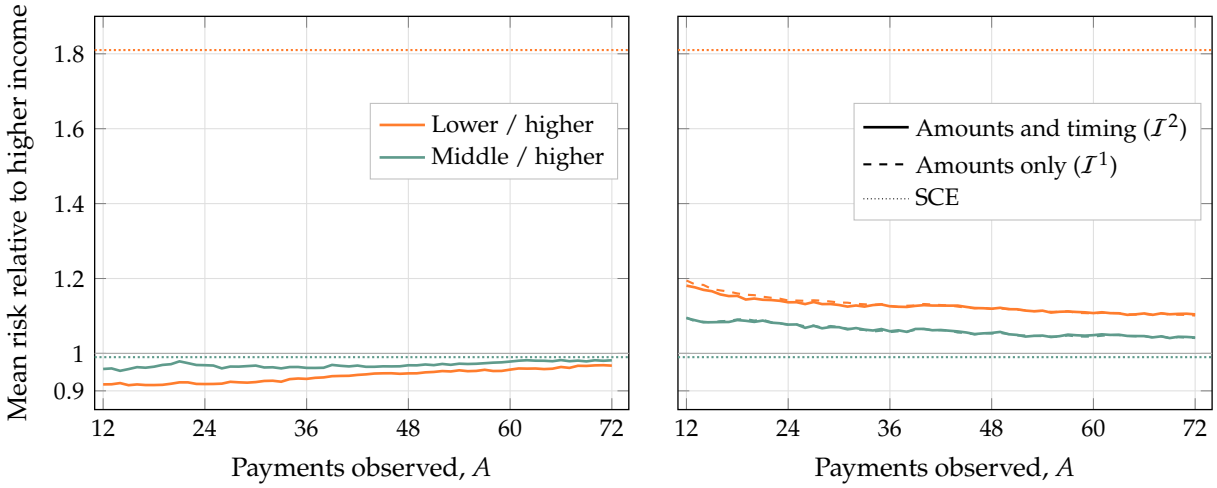
of the priors and the dependence between beliefs about bonus and other temporary variance (Appendix K.4).

amounts. This revised estimate enters successive forecasts until further payments change her assessment, so perceived risk can remain elevated even after earnings return to their previous level. However, under $\mathcal{I}^2$, she can assign annual-payment uncertainty to the scheduled month. A forecast made in that month includes uncertainty about next year's bonus; the following month's forecast concerns ordinary pay, so that contribution disappears even if her assessment of bonus variability is unchanged. Annual payments are more common and more variable at higher incomes, making these temporary increases a larger source of variation in their perceived risk. Knowing payment dates therefore reduces how strongly one month's risk assessment carries into the next, particularly at higher incomes, without requiring faster learning about the underlying earnings process.

The resulting lower–higher persistence gap under $\mathcal{I}^2$ ranges from 0.076 to 0.188 across learning histories of 12 to 72 payments, compared with 0.199 in the SCE. Payment timing thus reverses the income ordering under $\mathcal{I}^0$ and generates part of the observed gradient, although the fit remains imperfect: at $A = 48$, persistence is too high at middle and higher incomes and too low at lower incomes. Since persistence under $\mathcal{I}^2$ changes little after the first 48 payments, I retain $A = 48$ as the common learning history to limit the additional computational cost of simulating longer histories.

### 5.3.2 The Level of Perceived Earnings Risk

Figure 5 compares mean perceived risk across income groups relative to higher-income workers. Under total-pay information, $\mathcal{I}^0$ in panel (a), both ratios lie below one: higher-income workers perceive the most risk. Observing bonus amounts reverses this ordering in panel (b), where lower-income workers perceive more risk than middle- and higher-income workers. The lines under $\mathcal{I}^1$ and $\mathcal{I}^2$ nearly coincide. Recognising bonus amounts therefore matters for the income gradient in mean perceived risk, and knowing payment dates adds little to this effect.




(a) Total pay only

(b) Bonus pay observed

**Figure 5:** Relative perceived earnings risk by learning history

*Notes:* Each point is a ratio of group mean perceived risk immediately after payment $A$, with higher-income workers as the denominator. Income groups are reassigned at each $A$ using earnings over payments $A - 11$ to $A$. Dotted lines show the SCE ratios of 1.81 and 0.99; the thin grey line indicates equal mean risk. Appendix Figure K.2 shows the same plot but with corresponding levels.

The reversal reflects how workers interpret the earnings variation documented in Finding 1. Higher-income workers experience more bonus variation, and with only total pay observed, their estimates attribute part of it to permanent risk. Observing bonuses separately reduces their estimates of permanent risk more strongly. Workers remain uncertain about future bonus amounts, but expect less variation from permanent shocks, which accumulate over the twelve-month forecast horizon. This inference also depends on the prior over temporary pay composition: under the maintained prior, a large bonus can raise the worker's assessment of other temporary variance, leaving less of the variation in pay net of bonuses attributed to permanent shocks. Bonus recognition thus affects beliefs about future permanent risk as well as uncertainty about current permanent pay. This reduction in perceived risk from observing bonuses is consistent with Wang (2026), who finds that conventional risk estimates partly capture earnings variation workers already anticipate.

The resulting income differences nevertheless remain modest. At $A = 48$, observing bonus amounts raises the lower–higher mean-risk ratio from 0.95 to 1.11, compared with 1.81 in the SCE; the middle–higher ratio rises from 0.97 to 1.05, compared with 0.99 (see Appendix Table K.2 for the full results). The difficulty in matching the mean-risk ratios is that expected permanent risk dominates the twelve-month forecast, and the calibrated permanent variances are similar across income groups. Workers' estimates remain relatively close across the learning histories considered, so bonus recognition generates only modest income differences and changing $A$ provides little scope to close the remaining gap.[14]

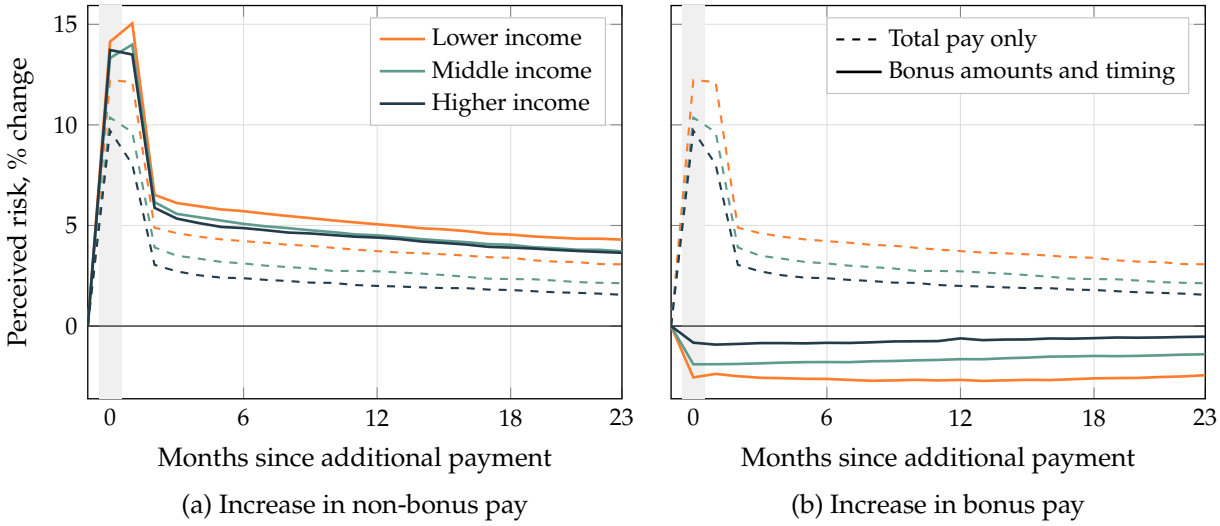
### 5.4 Model Applications

I now use the calibrated model to examine the implications of these findings in two practical settings. I first study how bonus recognition changes perceived earnings risk after a temporary pay increase, then consider a policy example to assess how transfers should be delivered to limit increases in perceived earnings risk.

#### 5.4.1 Responses to Temporary Payments

I consider a worker whose pay temporarily rises and examine what she learns from this increase about her future earnings risk. When its source is unclear, the payment may lead her to expect greater variation in permanent pay, whilst recognising it as a bonus allows her to attribute the increase and its subsequent disappearance to temporary pay instead. I reproduce this situation in two separate cases, adding a one-month increase of five log points to non-bonus pay $u_t$ in panel (a) and to bonus pay $b_t$ in panel (b). Figure 6 compares the resulting changes in perceived risk under $\mathcal{I}^0$ and $\mathcal{I}^2$, relative to the same earnings histories without the additional payment.

[14] A stronger reduction in perceived permanent risk could arise if workers subtract identified temporary payments but continue assigning the same share of the remaining variation to permanent shocks. The present model does not impose this restriction. Evidence of insufficient sensitivity to signal strength (Augenblick, Lazarus and Thaler, 2025) and cognitive uncertainty (Enke and Graeber, 2023) motivates considering it; these studies do not directly test this attribution rule for earnings risk.




(a) Increase in non-bonus pay

(b) Increase in bonus pay

**Figure 6:** Perceived risk after a temporary increase in pay

*Notes:* Simulation of 24,576 workers receiving a one-month increase of five log points after 48 payments (shaded month). Lines show percentage changes in mean twelve-month forecast variance relative to the same workers' histories without the increase. Income groups are fixed using earnings over the preceding twelve payments.

Under $\mathcal{I}^0$, the dashed responses coincide across panels: perceived risk rises by 10–12 percent and remains elevated after the payment ends. The worker interprets the rise and subsequent fall as evidence of greater permanent earnings variability, so returning to her previous pay does not restore her previous risk assessment. Under $\mathcal{I}^2$, observing ordinary bonuses separately does not resolve this uncertainty when the additional payment enters non-bonus pay. When the additional payment itself is identified as a bonus, however, subtracting it leaves the signal of permanent pay unchanged and the large positive response disappears. The small decline in perceived risk follows the inference about pay composition discussed in Section 5.3.2: under the maintained prior, a larger bonus can raise perceived temporary variability and reduce the variation attributed to permanent shocks. Because permanent shocks accumulate over the forecast horizon, this reduction outweighs the increase in temporary-pay risk in this calibration. The experiment therefore suggests that clearly identifying the source and temporary nature of a payment can limit its effect on perceived earnings risk without changing the amount workers receive.

### 5.4.2 Policy Application

Now in a policy application, consider a government transfer intended to stimulate the economy, such as a stimulus payment during the Covid pandemic. The previous experiment illustrated how a one-off receipt can affect perceived earnings risk when it is not distinguished from ordinary pay. I now extend this exercise to a longer-term recurring transfer, for instance an annual tax rebate. Using the calibrated model, I examine whether paying it all at once or spreading it equally across twelve months produces smaller changes in perceived earnings risk, and how recognising the transfer affects this comparison.

Each worker receives a fixed annual amount equal to 6 percent of her average monthly earnings over the preceding year. I pay this amount once a year or divide it into twelve equal instalments, keep the programme in place for three years, and follow beliefs for another year after it ends. For

each delivery schedule, both learners observe ordinary bonus amounts and dates under $\mathcal{I}^2$. One also identifies each government transfer and its amount when received; the other sees it only as part of total receipts.

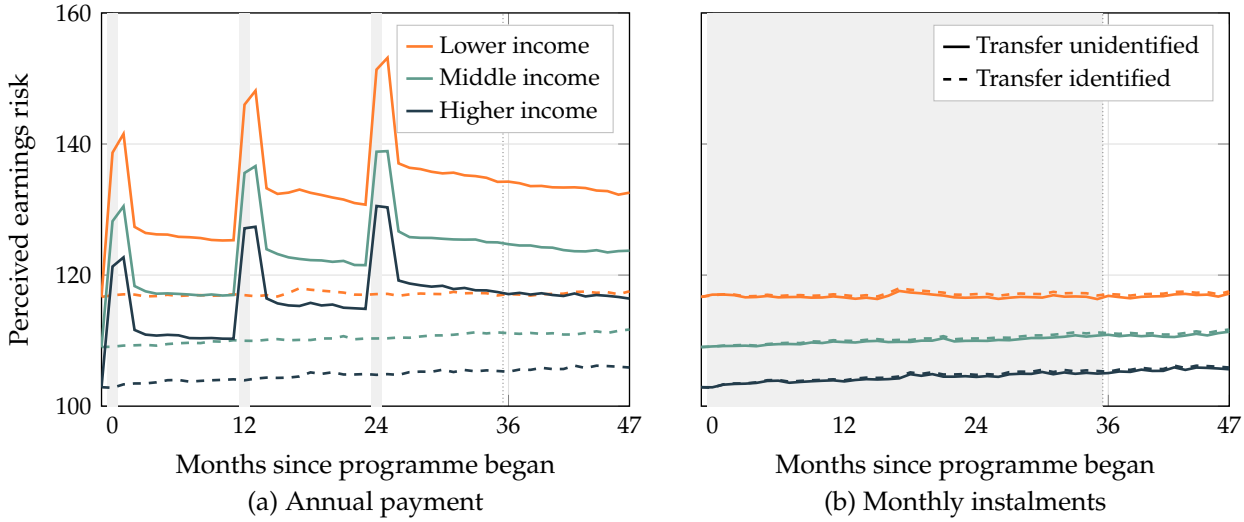



**Figure 7:** Transfer delivery and perceived earnings risk

*Notes:* Simulation of 24,576 workers. Lines show mean twelve-month earnings forecast variance in squared 100-log-point units. Both learners observe ordinary bonus amounts and dates ($\mathcal{I}^2$). Dashed lines identify the transfer at receipt and coincide with the no-transfer paths. Income groups update monthly using earnings over the preceding twelve payments, excluding transfers. Shading marks receipt; the dotted line marks programme end.

Figure 7 shows a large difference between the two delivery schedules when transfers are not identified (solid lines). The first annual payment in panel (a) raises mean perceived risk by 19–22 variance units and later receipts produce further increases. The worker reads the large receipt and the subsequent return to ordinary pay as evidence of greater earnings volatility. Monthly instalments in panel (b) produce much smaller revisions, with declines of less than one variance unit during the programme and no comparable spike at withdrawal. A fixed cash instalment also dampens proportional fluctuations in observed pay, consistent with these small negative responses.

The annual-payment effect persists after the programme ends. At the end of the follow-up year, perceived risk remains about 11–15 variance units above its no-transfer path, with the remaining response concentrated in beliefs about future permanent shocks. Recognising the transfer gives the worker an explanation for the movements in her receipts. With each receipt identified, she subtracts the government payment before learning about earnings, so perceived earnings risk follows the no-transfer path under either delivery schedule (dashed lines). These results favour clearly identifying the source and amount of government transfers, so recipients can distinguish support from changes in their earnings. Where recipients cannot make this distinction, monthly instalments limit the increase in perceived risk without reducing the annual amount they receive.

### 5.4.3 Macroeconomic Implications

I conclude the quantitative analysis by considering what these findings imply for macroeconomic consumption–saving models, focusing on precautionary motives, calibration and transfer design.

First, in heterogeneous-agent models with incomplete markets, households save to protect consumption against earnings losses they cannot fully insure (Aiyagari, 1994; Carroll, 1997). The desired buffer depends on the risk households perceive, so information about pay composition can affect both its size and its distribution across incomes. My model generates part of the income gradient in perceived risk from common initial beliefs, as workers learn from payments with different compositions. This provides an economic basis for heterogeneity in beliefs, which can help consumption models explain differences in wealth and spending (Carroll et al., 2017). Saving motives can then evolve with workers' payment histories. Greater perceived risk need not imply greater observed saving, however, since households may lack the resources to accumulate the protection they desire.

Second, learning can sustain a precautionary motive after the payment that prompted it has disappeared. In the temporary-payment experiment, an unidentified receipt raises the worker's assessment of permanent earnings variability, and returning to ordinary pay does not immediately reverse that assessment. For a given expected earnings path, the resulting desire to protect future consumption could therefore outlast the additional income. This connects to the precautionary saving generated by uncertainty about individual income components in Wang (2004). Predictable calendar changes in forecast risk need not involve such revisions to beliefs, meaning that a consumption model must therefore distinguish lasting changes in perceived earnings variability from changes in when uncertain payments occur.

Third, calibrating households' perceived risk directly from measured earnings variation can overstate the uncertainty they face. SIPP estimates used in models of precautionary saving and portfolio choice (Bayer et al., 2019; Choukhmane and de Silva, 2026), but workers may anticipate some variation that the econometrician treats as risk. Assigning that variation to their uncertainty can generate excessive precautionary buffers, contributing to the difficulty some consumption models face in matching low liquid wealth and large spending responses to additional income. Tao Wang (2026) shows that incorporating subjective expectations and risks improves both predictions. My analysis instead provides the micro-founded mechanism through which payment information generates a wedge between measured and perceived risk.

Lastly, transfer delivery can affect how households divide additional income between consumption and saving. An unlabelled transfer relaxes the household's budget constraint, but may also lead it to expect less stable earnings. The household may then save more of the additional income to protect future consumption. Monthly instalments instead induce smaller revisions to perceived risk, suggesting less scope for this precautionary response than an annual payment of the same entitlement. This provides one mechanism through which delivery could affect spending, although other effects may work in the opposite direction. Sahm et al. (2012), for instance, find a higher reported propensity to spend lump-sum payments than reduced withholding. My results therefore suggest that monthly delivery can limit revisions to perceived earnings risk without necessarily producing the largest spending response. That response also depends on households' liquidity needs, borrowing constraints and understanding of the programme. Quantifying these effects requires a consumption–saving model in which households respond to both the additional income and their revised beliefs, which I leave as an avenue for future research.

# 6 Conclusion

In this paper I show how information about pay composition can explain part of the difference between measured and perceived earnings risk across incomes. Empirically, I show that bonuses account for more temporary earnings variation and show stronger annual recurrence at higher incomes. In a learning model calibrated to these patterns, recognising bonuses generates a modest decline in mean perceived risk with income. Knowing payment dates helps reproduce the persistence gradient because annual-payment uncertainty enters and leaves successive forecasts, without requiring faster learning at higher incomes. The model nevertheless leaves much of the difference in mean risk unexplained. Transfer experiments show that unrecognised annual payments can raise perceived risk beyond the month of receipt, with much smaller effects when the same entitlement is paid monthly. These findings therefore suggest that models of consumption and saving should account for what households know about their payments when translating observed earnings variation into the risk they perceive, with potentially important quantitative implications for asset accumulation and spending responses.

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# A Measuring Perceived Earnings Risk

## A.1 Adjustment and Sample Comparisons

The adjustment separates differences associated with observed respondent characteristics and interview dates from the income-group comparison. I regress the bin-midpoint variance on age, age squared, sex, education, state, calendar year-month and interview-position indicators, using survey weights. Interview position counts consecutive monthly reports within a response spell, with positions twelve and above grouped together. The adjusted measure adds the weighted mean variance in the regression sample back to each residual. The comparisons below retain this adjustment when restricting the sample, so changes in the reported means reflect the respondents retained rather than a new regression.

Table A.1 compares raw and adjusted perceived variance. The lower–higher difference falls from 14.11 to 9.22 after adjustment. Among high-numeracy respondents, the raw difference remains positive, but the adjusted difference is close to zero. Among low-numeracy respondents, the adjusted difference is substantially larger.

**Table A.1:** Perceived variance and numeracy

| Measure and sample | Lower | Middle | Higher | Lower–higher difference (SE) |
|---|---|---|---|---|
| Raw, all respondents | 22.85 | 11.37 | 8.74 | 14.11 (1.49) |
| Adjusted, all respondents | 20.62 | 11.33 | 11.41 | 9.22 (1.38) |
| Raw, high numeracy | 11.71 | 8.90 | 8.28 | 3.43 (1.25) |
| Adjusted, high numeracy | 10.95 | 9.41 | 11.15 | −0.21 (1.13) |
| Raw, low numeracy | 35.28 | 18.43 | 12.65 | 22.63 (3.05) |
| Adjusted, low numeracy | 31.41 | 16.83 | 13.52 | 17.89 (2.87) |

*Notes:* Survey-weighted means in squared percentage points. Standard errors for lower–higher differences are in parentheses and cluster by respondent. Numeracy uses each respondent's first available SCE classification.

The high-numeracy sample contains 28,738 reports from 3,021 respondents, compared with 35,697 reports from 3,790 respondents in the full sample. Low-numeracy respondents account for 47.3 percent of weighted reports at lower incomes, 25.9 percent at middle incomes and 10.6 percent at higher incomes. Restricting the sample therefore changes its composition differently across income groups. This limits how far the high-numeracy comparison can describe the full population; it does not establish whether numeracy captures differences in earnings uncertainty or in how respondents express it. Appendix D.2 also examines persistence within the high-numeracy sample.

## A.2 Income Cut-offs

Table A.2 varies the household-income boundaries while holding the retained reports, survey weights and adjustment fixed. The adjusted gap remains positive when the lower-income group is narrowed to households below $20,000, the higher-income group is narrowed to those above $200,000, or both. It is much smaller when the lower comparison group instead includes all

households below $100,000. The contrast is therefore strongest when the comparison isolates the lowest-income households.

**Table A.2:** Perceived variance under alternative income cut-offs

| Household income | | Adjusted mean | | Difference (SE) | $p$ |
|---|---|---|---|---|---|
| Lower group | Higher group | Lower | Higher | | |
| $< \$50,000$ | $\geq \$100,000$ | 20.62 | 11.41 | 9.22 (1.38) | $< 0.001$ |
| $< \$20,000$ | $\geq \$100,000$ | 37.49 | 11.41 | 26.09 (3.74) | $< 0.001$ |
| $< \$50,000$ | $\geq \$200,000$ | 20.62 | 13.95 | 6.67 (1.54) | $< 0.001$ |
| $< \$20,000$ | $\geq \$200,000$ | 37.49 | 13.95 | 23.54 (3.81) | $< 0.001$ |
| $< \$100,000$ | $\geq \$200,000$ | 15.28 | 13.95 | 1.32 (1.05) | 0.207 |

*Notes:* Survey-weighted adjusted means in squared percentage points. Differences are lower minus higher, with respondent-clustered standard errors in parentheses. Cut-offs use the SCE income bands; the lowest band contains 102 respondents.

# B The Distribution of Pay Components

## B.1 Payment Size

Table B.1 supplements the mean payments in Table 1 with monthly receipt rates and quantiles. Positive bonuses are particularly dispersed. In the higher-income group, the median bonus is $2,030, compared with $21,000 at the ninetieth percentile and $168,000 at the ninety-ninth percentile. The mean of $10,150 therefore describes a distribution with a long upper tail. Bonuses also account for a substantial share of pay when received, averaging between 21.6 and 30.4 percent across the three income groups.

**Table B.1:** Monthly receipt and the distribution of positive payments

| | Receipt (%) | Median ($) | 90th percentile ($) | 99th percentile ($) | Share of pay when received (%) |
|---|---|---|---|---|---|
| *A. Lower income* | | | | | |
| Bonuses | 0.77 | 300 | 2,000 | 13,000 | 22.5 |
| Commissions | 2.48 | 1,100 | 3,200 | 14,700 | 76.5 |
| Tips | 2.56 | 400 | 1,500 | 4,180 | 38.5 |
| Overtime | 2.65 | 225 | 900 | 2,400 | 12.1 |
| *B. Middle income* | | | | | |
| Bonuses | 1.22 | 500 | 5,000 | 20,000 | 21.6 |
| Commissions | 2.90 | 1,800 | 5,000 | 14,000 | 67.1 |
| Tips | 1.60 | 540 | 2,000 | 6,230 | 36.7 |
| Overtime | 4.12 | 400 | 1,500 | 5,980 | 13.8 |
| *C. Higher income* | | | | | |
| Bonuses | 1.89 | 2,030 | 21,000 | 168,000 | 30.4 |
| Commissions | 3.45 | 3,220 | 14,300 | 59,500 | 59.6 |
| Tips | 1.12 | 500 | 2,800 | 8,540 | 35.3 |
| Overtime | 3.67 | 560 | 2,900 | 9,350 | 14.8 |

*Notes:* Survey-weighted SIPP estimates, 2013–2023; amounts in nominal dollars. Quantiles and mean pay shares condition on positive receipt. Hours may vary.

## B.2 Receipt within a Year

I measure annual receipt among workers observed for all twelve months of a calendar year in the same recorded job. The sample contains 168,313 worker-years from 92,481 workers. Each worker-year receives the average of its monthly survey weights. A year without receipt does not establish that a worker is ineligible for these payments.

Conditional on receiving a bonus during the year, workers usually receive it in only a few months. This is especially pronounced at higher incomes, where 71.2 percent of bonus recipients record exactly one payment month. These counts describe payment frequency; they do not establish whether workers knew the timing or size of a payment in advance.

# C The Earnings Benchmark

## C.1 Hours and Sample Period

The SIPP income groups use each worker's median annualised household income over her observed history. The benchmark retains monthly changes within the same job whether or not reported hours change. The lower-, middle- and higher-income samples contain 424,472, 702,110 and 881,152 changes, and 375,464, 623,762 and 779,812 adjacent pairs, respectively.

The unchanged-hours comparison requires the same weekly hours at both endpoints of a monthly change and across all three months of an adjacent pair. It retains 414,575, 690,502 and 869,519 changes. Almost all workers still contribute observations: the restriction removes one lower-income worker and no workers in the other groups. It primarily excludes particular months, whose contribution to squared earnings changes is disproportionately large. Figure C.1 shows the corresponding income gradient in reported hours variation.

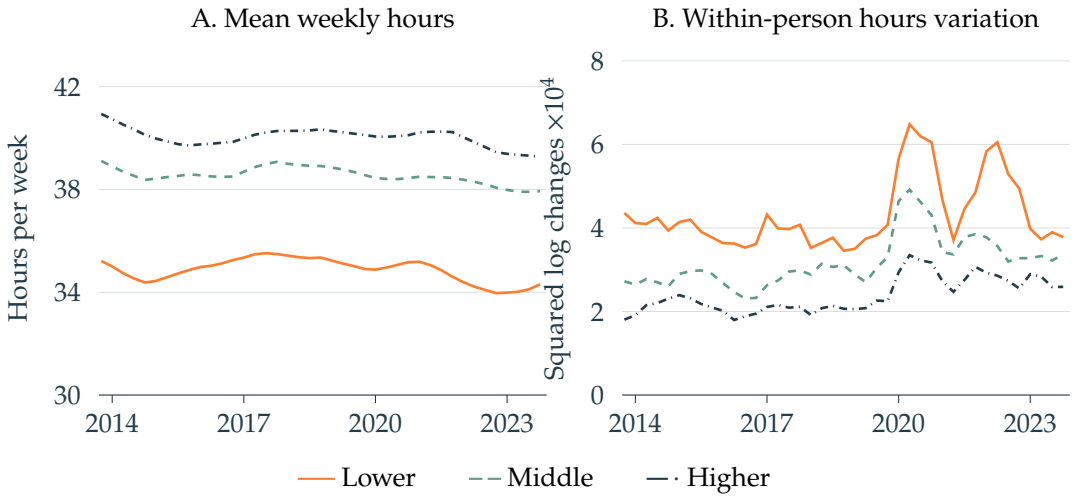



**Figure C.1:** Hours worked and hours variation by income group

*Notes:* SIPP, 2013–2023, same-job histories. Panels A and B show survey-weighted mean weekly hours and mean squared monthly changes in log hours, respectively; the latter are scaled by $10^4$. January changes are excluded and squared changes are trimmed at pooled first and ninety-ninth percentiles. Lines show four-quarter trailing moving averages of quarterly means.

The restriction matters most for the permanent component (Table C.1). Requiring unchanged hours lowers the permanent estimate in every group and produces an increasing income profile. The temporary component rises with income under either sample definition. Restricting the period to 2017–2023 raises both components relative to the full-period estimates in the unchanged-hours sample. These comparisons use the same estimation procedure, with residuals and pooled trimming thresholds re-estimated for each period. Within a period, the comparison across hours restrictions holds both fixed.

**Table C.1:** Earnings moments by hours restriction and sample period

| | Lower | | Middle | | Higher | |
|---|---|---|---|---|---|---|
| Sample | $\mathcal{P}$ | $\mathcal{T}$ | $\mathcal{P}$ | $\mathcal{T}$ | $\mathcal{P}$ | $\mathcal{T}$ |
| 2013–2023, hours may change | 8.93 | 2.05 | 8.64 | 3.06 | 8.39 | 4.06 |
| 2013–2023, unchanged hours | 4.03 | 2.24 | 4.68 | 3.23 | 5.13 | 4.22 |
| 2017–2023, unchanged hours | 6.08 | 2.83 | 6.31 | 4.72 | 7.09 | 6.08 |

*Notes:* Survey-weighted monthly log moments scaled by $10^4$. January changes are excluded; squared changes are trimmed at pooled first and ninety-ninth percentiles.

### C.2 Dependence beyond One Month

The benchmark imposes restrictions beyond the two moments used to recover $\mathcal{P}_g$ and $\mathcal{T}_g$. For mean-zero earnings changes, it predicts

$$\mathbb{E}[d_{it}d_{i,t-k} \mid g] = \begin{cases} -\mathcal{T}_g, & k = 1, \\ 0, & k \geq 2. \end{cases} \tag{C.1}$$

I estimate these products at lags one to six within the same job and annual reference period, allowing reported hours to change. Both endpoint changes must satisfy the main trimming rule. The first-lag estimates reproduce Table 2.

Table C.2 shows dependence beyond the first lag in all three income groups. The second-lag product is positive at lower and middle incomes, while the third-lag product is positive at higher incomes. The joint restriction that lags two through six are zero is rejected in each group. The simple process therefore does not reproduce the full pattern of retained earnings changes. These tests concern the trimmed, same-job sample; they do not separately identify earnings dynamics, reporting error and the effects of sample selection.

**Table C.2:** Products of earnings changes at different lags

| Lag in months | Lower | Middle | Higher |
|---|---|---|---|
| 1 | −2.046 (0.112) | −3.058 (0.130) | −4.060 (0.133) |
| 2 | 0.546 (0.071) | 0.375 (0.086) | −0.091 (0.090) |
| 3 | −0.343 (0.115) | 0.104 (0.131) | 0.942 (0.151) |
| 4 | 0.217 (0.084) | 0.048 (0.095) | −0.375 (0.091) |
| 5 | −0.171 (0.074) | −0.610 (0.099) | −0.801 (0.091) |
| 6 | −0.451 (0.107) | 0.183 (0.144) | 0.796 (0.151) |
| Joint $p$, lags 2–6 equal zero | $< 0.001$ | $< 0.001$ | $< 0.001$ |

*Notes:* Survey-weighted products, scaled by $10^4$, using available pairs at each lag. Household-clustered standard errors condition on fitted residuals and trimming. Joint tests have five restrictions.

### C.3 Annual Reporting Boundaries

December and January earnings are reported in separate annual interviews. Differences in recall or reporting across interviews can therefore enter the recorded change between these months.

Table C.3 compares December-to-January changes with other monthly changes in the same job, allowing hours to change. January changes are much larger in every income group. Applying the main sample's trimming thresholds to both sets of changes reduces their levels substantially but leaves a large difference. Actual seasonal payments may also contribute to this pattern, so the comparison does not attribute the entire January difference to reporting error.

**Table C.3:** Earnings changes at the annual reporting boundary

| | No trimming | | Main trimming thresholds | |
|---|---|---|---|---|
| Income group | February–December | January | February–December | January |
| Lower | 108.46 | 5,225.05 | 13.02 | 306.21 |
| Middle | 137.57 | 4,810.58 | 14.75 | 309.76 |
| Higher | 245.26 | 5,148.33 | 16.51 | 289.42 |

*Notes:* Mean squared adjusted log hourly earnings changes, scaled by $10^4$, allowing reported hours to change. Month refers to the later endpoint. Both trimmed columns use the main non-January sample's pooled thresholds.

The twelve-month benchmark in the main text extrapolates from non-January monthly moments. More generally, for a stationary sequence of monthly changes with autocovariances $\gamma_{k,g}$,

$$\operatorname{Var}(z_{i,t+H} - z_{it} \mid g) = H\gamma_{0,g} + 2\sum_{k=1}^{H-1}(H-k)\gamma_{k,g}. \tag{C.2}$$

Setting $\gamma_{0,g} = \mathcal{P}_g + 2\mathcal{T}_g$, $\gamma_{1,g} = -\mathcal{T}_g$ and the remaining autocovariances to zero gives $H\mathcal{P}_g + 2\mathcal{T}_g$. Dependence at longer lags can change this extrapolation. A direct twelve-month difference also crosses an annual reporting boundary, even when neither endpoint is January. The monthly benchmark and a directly measured twelve-month change therefore differ in their exposure to the survey seam as well as in their assumptions about earnings dynamics.

# D Persistence and Statistical Inference

## D.1 The Persistence Specification

For each income group, I estimate

$$q_{it} = a_i + \phi_g q_{i,t-1} + \varepsilon_{it}, \tag{D.1}$$

where $q_{it}$ denotes adjusted perceived variance in the SCE or the squared earnings residual used as the SIPP volatility proxy. The unit effect $a_i$ refers to an uninterrupted response spell in the SCE and a job spell in the SIPP. Hours may change within the SIPP spell; lags never connect observations from different jobs.

The SIPP proxy uses adjacent, retained monthly earnings changes. I regress the current change on a constant and its one-month lag, pooling the income groups and using monthly survey weights. The fitted lag coefficient is $-0.2588$. Squaring the residual gives the proxy; I apply no further trimming after this step. There are 375,464, 623,762 and 779,812 proxy observations across the lower-, middle- and higher-income groups. Missing months remain gaps when constructing subsequent lags.

System GMM combines the equation in first differences with the equation in levels. Differencing removes the unit effect, but the lagged regressor $\Delta q_{i,t-1}$ and the differenced error $\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$ both contain the previous month's shock, $\varepsilon_{i,t-1}$. I therefore instrument the lagged regressor with earlier levels of $q_{it}$. The second lag is valid if the level errors are serially uncorrelated; serial correlation can require instruments further in the past.

The main specification uses $q_{i,t-3}$ and $q_{i,t-4}$ in the differenced equation, leaving an additional month of separation relative to the second lag. Their validity still depends on the error process. The level equation uses $\Delta q_{i,t-1}$ as its instrument and requires a separate restriction. The identifying moment restrictions are

$$\mathbb{E}[q_{i,t-j}\,\Delta\varepsilon_{it}] = 0, \qquad j \in \{3, 4\}, \tag{D.2}$$

$$\mathbb{E}[\Delta q_{i,t-1}(a_i + \varepsilon_{it})] = 0. \tag{D.3}$$

The level restriction additionally requires changes in the outcome to be uncorrelated with the unit effect. I collapse the instruments into one column per lag and use two-step estimation. Lags refer to calendar months and stay within the relevant spell or segment. The GMM calculation holds the first available survey weight within each unit fixed over its history.

## D.2 Alternative Instruments and Numeracy

Table D.1 moves the instrument window one month earlier or later. Perceived risk remains more persistent at lower than at higher incomes, whereas the SIPP proxy remains more persistent at higher incomes. The SCE ordering also appears in raw perceived variance and among high-numeracy respondents. Thus, the near-zero adjusted level gap in that subsample does not extend to its persistence estimates.

The instrument restrictions do not pass every specification test (Table D.2). In the main specifica-

**Table D.1:** Sensitivity of the persistence estimates

| Specification | Lower | Middle | Higher |
|---|---|---|---|
| *A. SCE perceived variance* | | | |
| Adjusted, instruments at lags 2–3 | 0.346 (0.044) | 0.108 (0.032) | 0.139 (0.022) |
| Adjusted, lags 3–4 (main) | 0.343 (0.046) | 0.142 (0.026) | 0.144 (0.025) |
| Adjusted, lags 4–5 | 0.350 (0.044) | 0.105 (0.038) | 0.112 (0.034) |
| Raw variance, lags 3–4 | 0.355 (0.042) | 0.128 (0.028) | 0.124 (0.034) |
| Adjusted, high numeracy, lags 3–4 | 0.274 (0.075) | 0.102 (0.043) | 0.126 (0.037) |
| *B. SIPP volatility proxy* | | | |
| Instruments at lags 2–3 | 0.193 (0.012) | 0.206 (0.010) | 0.275 (0.009) |
| Lags 3–4 (main) | 0.193 (0.013) | 0.187 (0.010) | 0.235 (0.009) |
| Lags 4–5 | 0.237 (0.015) | 0.239 (0.012) | 0.279 (0.010) |

*Notes:* Monthly AR(1) coefficients from two-step system GMM. Standard errors in parentheses cluster by response spell and month (SCE) or job spell (SIPP). Alternative instrument windows change only the differenced-equation instruments.

tion, the Hansen test rejects for the middle-income SCE group and for all three SIPP groups. The persistence comparisons should therefore be read alongside these rejections. A stable ordering across instrument windows does not validate the AR(1) specification or identify the persistence of latent earnings risk.

**Table D.2:** Hansen tests of the GMM moment restrictions

| Specification | Lower | Middle | Higher |
|---|---|---|---|
| *A. SCE perceived variance* | | | |
| Adjusted, instruments at lags 2–3 | 0.275 | 0.041 | 0.059 |
| Adjusted, lags 3–4 (main) | 0.261 | 0.005 | 0.063 |
| Adjusted, lags 4–5 | 0.398 | 0.126 | 0.279 |
| Raw variance, lags 3–4 | 0.195 | 0.013 | 0.145 |
| Adjusted, high numeracy, lags 3–4 | 0.789 | 0.019 | 0.016 |
| *B. SIPP volatility proxy* | | | |
| Instruments at lags 2–3 | $< 0.001$ | $< 0.001$ | $< 0.001$ |
| Lags 3–4 (main) | $< 0.001$ | $< 0.001$ | $< 0.001$ |
| Lags 4–5 | 0.226 | $< 0.001$ | $< 0.001$ |

*Notes:* $p$-values for the two overidentifying restrictions in each specification. Tests use the unit-clustered GMM weighting matrix.

### D.3 Standard Errors and Income-Group Comparisons

For SCE levels, I obtain group means and their covariance from a survey-weighted regression on income-group indicators, clustering by respondent. For the SIPP moments in Table 2, I calculate analytic standard errors clustered by household. The calculation retains the covariance between the estimated second moment and adjacent-change product when constructing the permanent, transitory and twelve-month variances. Fitted earnings residuals, group assignments, sample membership and trimming thresholds are held fixed.

For persistence, the SCE covariance clusters by response spell and calendar month, subtracting their intersection. The comparison across income groups retains the covariance induced by shared months. SIPP standard errors cluster by job spell; these clusters do not overlap across income groups. The persistence standard errors condition on the fitted GMM weighting matrix and, for SIPP, the preliminary earnings regression.

The Wald tests in Tables 2 and 3 test equality between higher- and lower-income estimates. For a generic estimate $\widehat{b}_g$, the statistic is

$$W = \frac{(\widehat{b}_{\text{Higher}} - \widehat{b}_{\text{Lower}})^2}{\widehat{\text{Var}}(\widehat{b}_{\text{Higher}}) + \widehat{\text{Var}}(\widehat{b}_{\text{Lower}}) - 2\widehat{\text{Cov}}(\widehat{b}_{\text{Higher}}, \widehat{b}_{\text{Lower}})}. \tag{D.4}$$

I compare this statistic with a chi-squared distribution with one degree of freedom. It tests the higher–lower difference, rather than equality across all three groups or equality between the two surveys. These standard errors do not use Census replicate weights or incorporate the estimation of the preliminary adjustments.

# E Accounting for Variation across Pay Components

## E.1 Additive Contributions to Log Earnings Changes

Let $W_{it}$ denote total recorded pay and $W_{it}^k$ component $k$. I allocate adjusted pay using their shares, $s_{it}^k = W_{it}^k / W_{it}$:

$$\widetilde{W}_{it} = \exp(z_{it}), \qquad \widetilde{W}_{it}^k = s_{it}^k \widetilde{W}_{it}, \qquad \sum_k \widetilde{W}_{it}^k = \widetilde{W}_{it}. \tag{E.1}$$

For positive totals, their logarithmic mean is

$$M(a,b) = \frac{a-b}{\log a - \log b}, \qquad M(a,a) = a. \tag{E.2}$$

Component $k$ contributes

$$d_{it}^k = \frac{\widetilde{W}_{it}^k - \widetilde{W}_{i,t-1}^k}{M(\widetilde{W}_{it}, \widetilde{W}_{i,t-1})}. \tag{E.3}$$

These contributions sum exactly to $z_{it} - z_{i,t-1}$. Only total pay enters a logarithm, so zero components and a negative measured remainder remain admissible.

I use the benchmark's monthly changes, adjacent pairs and survey weights, with a common sample across components for each moment. Centring the adjacent products changes the bonus shares only slightly, from 35.7, 53.0 and 67.5 percent to 35.5, 52.9 and 67.3 percent.

Household-clustered standard errors account for covariance with the group total and across income groups, holding fitted residuals, the sample and trimming thresholds fixed. The higher–lower bonus-share difference is 31.77 percentage points (SE 3.18).

## E.2 Contributions to Permanent-Innovation Variation

Writing $\mathcal{S}_{kg} = \mathbb{E}[d_{it}^k d_{it} \mid g]$, the permanent contribution is

$$\mathcal{P}_{kg} = \mathcal{S}_{kg} - 2\mathcal{T}_{kg}, \qquad \sum_k \mathcal{P}_{kg} = \mathcal{P}_g. \tag{E.4}$$

Wages and salaries excluding itemised payments account for 81.8–86.2 percent of permanent-innovation variation (Figure E.1). The bonus share rises from 4.7 to 12.4 percent across income groups.

Each contribution subtracts twice the corresponding transitory allocation from its contribution to the second moment of earnings changes. These accounting contributions do not estimate separate shock variances for each category.

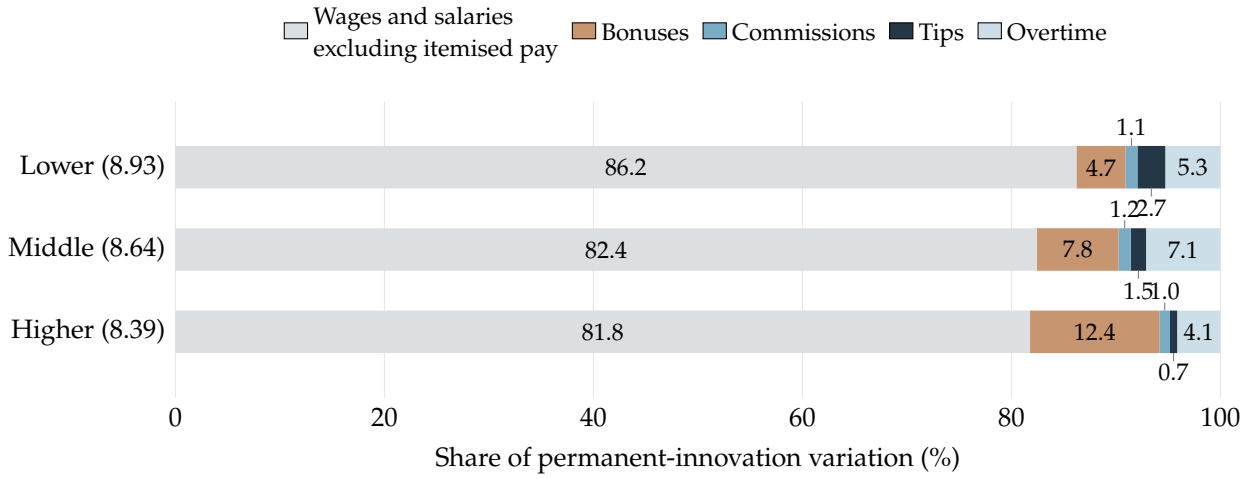



**Figure E.1:** Pay components' shares of permanent-innovation variation

*Notes:* SIPP, 2013–2023, using the hours-may-change sample of Table 2. Shares include cross-category terms. Group totals appear in parentheses, with log moments scaled by $10^4$.

# F Payment Duration

## F.1 Specification

The payment profiles use months in which the worker received none of the relevant component in either of the preceding two months. Current payments may be positive or zero, and future receipt is unrestricted. Requiring two prior zero months excludes ongoing monthly payments while retaining payments that recur every quarter. This sample therefore describes payments following a gap rather than all recipients of a component.

The controls $C_{it}$ in equation (5) include twelve lags of total hourly pay, the component's hourly amount and receipt indicator, and the combined hourly amount of the other itemised payments. Amounts use the same denominator as the dependent variable. I also control for log average prior hourly earnings and log hours in the preceding month, alongside calendar-date and income-group effects. Regressions use survey weights and standard errors clustered by worker. Since the current payment is also the dependent variable at $h = 0$, $\beta_0^k = 1$ by construction.

All horizons use the same valid, non-imputed observations on the same first recorded job from $t-12$ through $t+6$. January initial months are excluded, and hours may change. I remove positive normalised initial payments above the within-income-group 99th percentile. The main sample contains 3,499 positive bonus payments from 2,192 workers and 514 positive overtime payments from 380 workers, alongside the zero-payment observations. Separate income-group estimates are particularly limited for commissions: the lower-income sample contains only seven positive payments from five workers.

## F.2 Alternative Payment Histories and Other Components

The contrast between bonuses and overtime remains when the required history without receipt changes. Table F.1 reports selected coefficients using one, two or three prior zero months. The bonus coefficient falls close to zero immediately in each sample, whereas the overtime coefficient

remains positive through month six. The estimated size of the later overtime payments varies across samples.

**Table F.1:** Payment profiles under alternative receipt histories

| Component | Prior zero months | Month 1 | Month 3 | Month 6 |
|---|---|---|---|---|
| Bonuses | 1 | −0.011 (0.008) | 0.035 (0.009) | 0.045 (0.011) |
| | 2 | −0.010 (0.008) | 0.034 (0.009) | 0.044 (0.011) |
| | 3 | −0.010 (0.008) | 0.015 (0.007) | 0.032 (0.010) |
| Overtime | 1 | 0.564 (0.076) | 0.335 (0.088) | 0.265 (0.102) |
| | 2 | 0.579 (0.082) | 0.329 (0.092) | 0.251 (0.104) |
| | 3 | 0.620 (0.086) | 0.260 (0.078) | 0.118 (0.036) |
| Commissions | 2 | −0.025 (0.016) | 0.449 (0.089) | 0.392 (0.105) |

*Notes:* Coefficients from equation (5). Worker-clustered standard errors in parentheses. The upper-1% trimming threshold is recalculated within each income group and receipt-history sample.

The overtime estimates are sensitive to large initial payments. Without trimming, the two-month-history coefficient is 0.21 at month one and 0.07 at month six, compared with 0.58 and 0.25 in the main specification. One worker accounts for nearly half the identifying variation in the untrimmed regressor. Holding the three-month sample's trimming thresholds fixed across receipt histories preserves the continued overtime payments; for the two-month history, the month-six coefficient is 0.24.

Commissions display peaks at months three and six, consistent with quarterly recurrence. The two-month-history sample contains 248 positive payments from 114 workers. Requiring three prior zero months excludes regular quarterly payments and weakens the month-three coefficient. For tips, only 11 positive payments from seven workers remain under the main rule, providing too little information for a reliable duration estimate.

### F.3 Twelve-Month Payment Profiles

I extend equation (5) through month twelve, re-estimating every horizon on a common sample within each component. The sample requires valid, non-imputed records in the same job from $t - 12$ through $t + 12$, with the same controls, weights and trimming rule as before. The longer observation window reduces the sample, so the first six months can differ from Figure 3.

Figure F.1 shows a pronounced annual bonus peak, with a month-twelve coefficient of 0.433 (SE 0.041). The bonus sample contains 2,133 positive starts from 1,329 workers. The commission and overtime profiles rely on relatively few positive starts: 141 from 54 workers for commissions and 331 from 242 workers for overtime, whose later coefficients are imprecisely estimated.

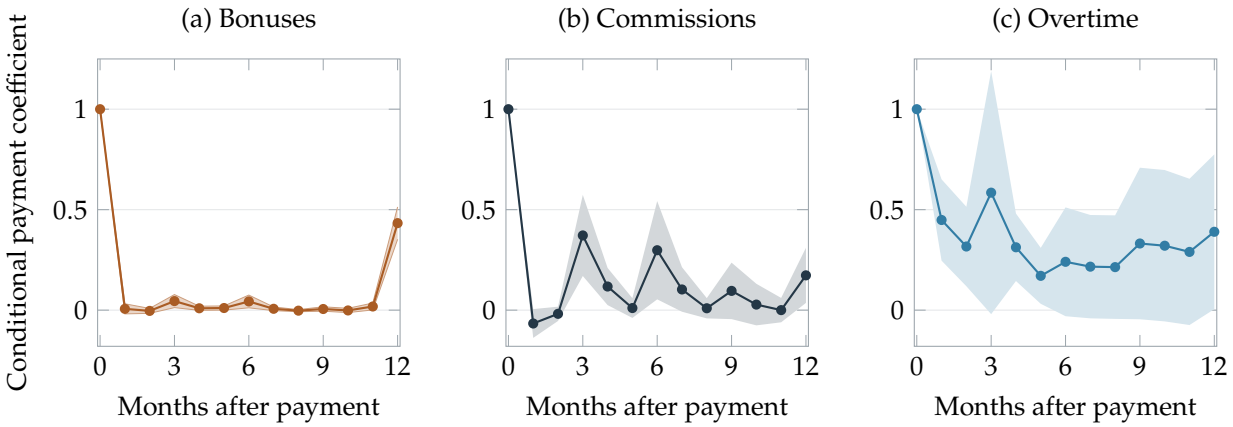



**Figure F.1:** Payment profiles through twelve months

*Notes:* SIPP, 2013–2023, pooled income groups. Coefficients relate future to current normalised hourly payments; the initial coefficient is one by construction. Shading gives 95% pointwise confidence intervals with standard errors clustered by worker.

# G Annual Recurrence in Pay Components

## G.1 Specification and Inference

For each component $k$, let $Y_{it}^k$ denote either an indicator for a positive payment or the monthly payment amount in nominal dollars. Define the preceding eleven-month average as $\bar{Y}_{it}^k = \frac{1}{11}\sum_{s=1}^{11} Y_{i,t-s}^k$. Within each income group $g$, I apply equation (6) to component $k$. The fixed effects absorb stable differences across worker–job matches and common calendar month–year effects. Last month's payment is included separately from the eleven-month average to allow it to carry additional information. The annual coefficient $\gamma_{kg}$ measures the association with the same-month payment a year earlier, conditional on these controls. Multiplying this coefficient by 100 gives percentage points per prior receipt in the indicator regression, or dollars per \$100 more last year in the amount regression.

The sample requires positive total earnings and survey weights, observed amounts for all four itemised components, and a complete preceding twelve-month history on the same first recorded job. January outcomes are excluded, and hours may change. Amount regressions include zero-payment months and use the recorded nominal dollar amounts without additional trimming or rescaling. All four components use the same observations within each income group: 142,237 at lower incomes, 255,594 at middle incomes and 321,742 at higher incomes.

Regressions use survey weights and standard errors clustered by worker. The higher–lower comparisons test equality of the two annual coefficients using a joint worker-clustered covariance matrix, which preserves dependence across equations. Each reported $p$-value is a separate, unadjusted one-restriction Wald test. These descriptive regressions do not directly measure workers' information or forecast accuracy; dynamic fixed effects, finite histories and survey reporting may affect the coefficients.

### G.2 Comparison across Components

Table G.1 applies the same specification to bonuses, commissions, tips and overtime. Only bonuses have positive annual coefficients after controlling for recent payment history. For the other categories, the negative annual coefficients concern the additional association with last year's payment, conditional on recent receipts or amounts. They do not imply that these payments are generally unpredictable. In particular, this annual comparison does not rule out the quarterly commission recurrence documented in Section 3.2.

**Table G.1:** Annual recurrence across pay components

| Component | Lower | Middle | Higher | Higher–lower $p$-value |
|---|---|---|---|---|
| *A. Receipt probability (pp) per receipt last year* | | | | |
| Bonuses | 6.90 (3.40) | 7.52 (1.78) | 20.59 (1.34) | $< 0.001$ |
| Commissions | −5.40 (1.59) | −5.50 (1.35) | −7.95 (1.56) | 0.253 |
| Tips | −3.76 (1.73) | −2.28 (0.78) | −3.68 (1.20) | 0.970 |
| Overtime | −5.62 (1.04) | −6.94 (0.68) | −6.96 (0.61) | 0.266 |
| *B. Amount ($) per $100 more last year* | | | | |
| Bonuses | 2.75 (2.29) | 7.34 (5.61) | 29.83 (3.22) | $< 0.001$ |
| Commissions | −9.35 (5.87) | −9.56 (3.51) | −19.43 (3.33) | 0.136 |
| Tips | −0.24 (0.16) | −2.30 (2.08) | −12.72 (4.83) | 0.010 |
| Overtime | −2.00 (0.90) | −3.98 (1.73) | −6.19 (1.19) | 0.005 |

*Notes:* Annual coefficients from equation (6), scaled by 100. Worker-clustered standard errors in parentheses. Wald tests compare higher- and lower-income coefficients.

### G.3 Recurrence at Different Lags

To assess whether the higher-income advantage is specific to annual recurrence, I repeat the bonus regressions at three, six and twelve months. Each regression uses exactly the same observations as the annual specification. Let $Y_{it}$ denote either bonus receipt or its monthly nominal amount, including zeros. For each lag $h \in \{3, 6, 12\}$ and income group $g$, I estimate

$$Y_{it} = c^{(h)}_{i,j(i,t)} + \tau^{(h)}_t + \gamma_{gh} Y_{i,t-h} + \eta_{gh} Y_{i,t-1} + \zeta_{gh} \bar{Y}^{(h)}_{it} + e^{(h)}_{it}, \tag{G.1}$$

where $\bar{Y}^{(h)}_{it} = \frac{1}{h-1} \sum_{s=1}^{h-1} Y_{i,t-s}$ averages payments over the preceding two, five or eleven months. The fixed effects, survey weights and worker clustering follow the annual specification. Table G.2 reports the receipt and amount estimates.

Receipt coefficients are positive in every income group at all three lags. The higher–lower difference is negative at three months ($p = 0.020$), not statistically significant at six months ($p = 0.436$), and positive at twelve months ($p < 0.001$). The positive income gradient is therefore specific to annual recurrence among these horizons.

Because the history average changes with $h$, the separate regressions condition on different payment histories. I therefore also include the three-, six- and twelve-month lags together in a single receipt regression. Recent-history controls are last month's receipt and average receipt over the nine remaining lags in the preceding year: months one, two, four, five, seven, eight, nine, ten and

**Table G.2:** Bonus recurrence at three, six and twelve months

| Lag | Lower | Middle | Higher | Higher–lower $p$-value |
|---|---|---|---|---|
| *A. Receipt probability (pp) per receipt at the indicated lag* | | | | |
| 3 months | 21.18 (2.70) | 19.46 (1.72) | 14.37 (1.10) | 0.020 |
| 6 months | 9.79 (2.12) | 10.82 (1.29) | 8.01 (0.84) | 0.436 |
| 12 months | 6.90 (3.40) | 7.52 (1.78) | 20.59 (1.34) | $< 0.001$ |
| *B. Amount ($) per $100 more at the indicated lag* | | | | |
| 3 months | 2.41 (2.79) | −0.40 (0.91) | −3.51 (1.47) | 0.061 |
| 6 months | −1.75 (2.24) | 3.36 (5.43) | −6.17 (1.31) | 0.089 |
| 12 months | 2.75 (2.29) | 7.34 (5.61) | 29.83 (3.22) | $< 0.001$ |

*Notes:* Each row uses a separate regression on the same sample. Worker-clustered standard errors in parentheses. Wald tests compare higher- and lower-income coefficients. Amounts include zero-payment months.

eleven. The fixed effects, sample, weights and clustering remain the same. This comparison holds the set of regressors fixed across the three lag coefficients.

Table G.3 reports these joint estimates. The higher-income receipt coefficient exceeds the lower-income coefficient only at twelve months, as in the separate regressions.

**Table G.3:** Bonus receipt recurrence with three lags included jointly

| Lag | Lower | Middle | Higher | Higher–lower $p$-value |
|---|---|---|---|---|
| 3 months | 20.62 (2.54) | 18.35 (1.52) | 13.90 (1.01) | 0.014 |
| 6 months | 9.14 (1.69) | 9.91 (0.98) | 5.99 (0.62) | 0.080 |
| 12 months | 5.97 (2.97) | 6.75 (1.55) | 20.13 (1.24) | $< 0.001$ |

*Notes:* Coefficients are in percentage points, with worker-clustered standard errors in parentheses. Wald tests compare higher- and lower-income coefficients.

### G.4 Bonus Timing and Temporary Earnings Variation

Table 5 applies the component accounting in Section 3.1 to months with bonus receipt in the same job one year earlier. For each pair of adjacent adjusted earnings changes, I calculate the bonus contribution

$$c_{it}^{b} = -\tfrac{1}{2}\left(d_{it}^{b} d_{i,t-1} + d_{it} d_{i,t-1}^{b}\right).$$

I attribute this contribution to the middle month $t-1$ and record whether the worker received a bonus twelve months before that month. I retain the benchmark trimming and reporting-seam exclusions, allow hours to change and restrict the sample to observations with the required same-job history. I also require enough history to determine receipt within one month of the prior-year date, keeping the sample common to the exact-date and wider-window calculations.

The first row reports the survey-weighted fraction of eligible observations with prior-year receipt. For the remaining rows, I multiply $c_{it}^{b}$ by the prior-receipt indicator and average over all eligible observations. I divide this average by either the full bonus contribution or total temporary

variation in the same restricted sample. These denominators are the survey-weighted means of $c_{it}^{b}$ and $-d_{it}d_{i,t-1}$, respectively. The resulting shares describe where measured variation is concentrated; they do not measure how much workers anticipated.

I calculate analytical standard errors by the delta method, accounting jointly for uncertainty in each ratio's numerator and denominator and clustering observations by worker. Wald tests use the joint covariance of the group estimates to test higher–lower differences. Inference conditions on the survey weights, prepared earnings adjustments, trimming and income classifications.

# H Forecasting Earnings with Bonus Information

### H.1 Specification and Evaluation

The forecasts in equation (7) use log total hourly earnings, including all pay components. The baseline contains current earnings, three exact monthly lags, and the arithmetic mean of all available earlier log hourly earnings in the same recorded job. Calendar controls are separate month-of-year and year effects. The bonus regression adds the current bonus share, its three exact monthly lags and the arithmetic mean of all available earlier monthly bonus shares in that job. Historical means exclude the current month. These means summarise older records; the specification does not include a separate twelve-month bonus lag.

I require positive total pay and hourly earnings, valid survey weights, and observed bonus and commission amounts. Commission records determine eligibility so that the comparison with additional commission information uses the same sample; they do not enter the bonus-only regressions. Workers must have three consecutive prior monthly observations and remain observed in the same recorded job through the forecast endpoint. The sample includes months without bonuses and allows hours to change. The twelve-month forecasts necessarily compare records from different annual reference periods.

I estimate each regression by survey-weighted least squares, separately by income group and horizon. A fixed allocation of workers to five sets is used throughout: all observations for a worker belong to the same set, and no worker contributes to both estimation and evaluation in a given split. Each specification uses the same observations, weights and splits. The income groups retain their classification from the full observed panel, and the fitted coefficients and calendar effects use other workers throughout the sample period. The exercise therefore compares predictive information in the records retrospectively.

I pool the predictions for excluded workers to calculate survey-weighted mean squared errors. Standard errors cluster by worker and use the delta method for the ratio of the two mean squared errors, allowing for their covariance. They condition on the fitted forecasts and do not include uncertainty from re-estimating the regressions. Higher–lower Wald tests use the corresponding standard error of the difference in percentage gains. Negative gains can arise when additional regressors worsen predictions on excluded workers; they do not imply that an optimal forecast becomes less accurate with more information.

### H.2 Current Information, Payment History and Additional Comparisons

Current bonus information accounts for most of the near-term forecast gain. Panel A of Table H.1 adds only past bonus shares to the total-pay baseline. Panel B then adds the current share. At higher incomes, past records alone reduce three-month mean squared error by 0.18 percent, while adding the current share reduces the remaining error by 3.65 percent. Each percentage uses the error of its own baseline, so the two percentages should not be added. At twelve months, the smaller overall gain comes mainly from past bonus records.

Panel C adds current and past commission shares alongside the bonus variables, using the same total-pay baseline, observations, weights and splits. The gains change little. Panel D evaluates the bonus forecasts only on observations with all three future endpoints available. I retain the forecasts estimated on each horizon's full sample and restrict their evaluation, without re-estimation. Both the larger gains at higher incomes and the decline with the horizon remain.

**Table H.1:** Forecast gains under alternative information sets and evaluation samples

| Forecast horizon | Lower | Middle | Higher |
|---|---|---|---|
| *A. Past bonus shares added to total-pay information* | | | |
| 3 months | 0.00 (0.01) | 0.02 (0.01) | 0.18 (0.02) |
| 6 months | −0.01 (0.01) | 0.04 (0.02) | 0.14 (0.03) |
| 12 months | −0.01 (0.01) | −0.05 (0.04) | 0.19 (0.06) |
| *B. Current bonus share added after past bonus shares* | | | |
| 3 months | 0.40 (0.12) | 0.94 (0.12) | 3.65 (0.17) |
| 6 months | 0.15 (0.06) | 0.34 (0.06) | 1.67 (0.10) |
| 12 months | 0.00 (0.00) | 0.02 (0.01) | 0.03 (0.01) |
| *C. Bonus and commission shares added to total-pay information* | | | |
| 3 months | 0.39 (0.12) | 0.99 (0.13) | 3.88 (0.18) |
| 6 months | 0.14 (0.08) | 0.41 (0.06) | 1.83 (0.11) |
| 12 months | −0.09 (0.29) | −0.02 (0.07) | 0.21 (0.08) |
| *D. Bonus information, common evaluation observations across horizons* | | | |
| 3 months | 0.34 (0.08) | 0.62 (0.14) | 2.59 (0.18) |
| 6 months | 0.14 (0.04) | 0.22 (0.06) | 1.18 (0.11) |
| 12 months | −0.01 (0.01) | −0.04 (0.04) | 0.21 (0.06) |

*Notes:* Percentage reductions in survey-weighted mean squared forecast error. Approximate worker-clustered standard errors in parentheses condition on fitted forecasts. Panels A–C use horizon-specific samples; panel D uses a common evaluation sample across horizons.

### H.3 Twelve-Month Forecasts with All Pay Components

I extend the twelve-month forecast comparison to include all four itemised pay components. For each additional component, I add its current share of total pay, three monthly lags and its average share over all earlier observations in the same job. Table H.2 compares bonus information with information on all four components. The bonus-and-commission comparison appears in Table H.1, panel C. Every regression uses the same total-pay baseline, observations, survey weights and worker splits. The additional components require no reduction in the sample.

At higher incomes, the forecast gain rises from 0.21 percent with bonus information to 0.43 percent

**Table H.2:** Twelve-month forecast gains from additional pay components

| Information added | Lower | Middle | Higher |
|---|---|---|---|
| *A. Reduction in mean squared forecast error (%)* | | | |
| Bonuses | −0.01 (0.01) | −0.04 (0.04) | 0.21 (0.06) |
| All four components | −0.08 (0.31) | −0.05 (0.10) | 0.43 (0.12) |
| *B. Additional gain from all four components over bonuses alone* | | | |
| Gain (pp) | −0.07 (0.31) | −0.01 (0.09) | 0.22 (0.10) |
| Paired $p$-value | 0.820 | 0.877 | 0.025 |

*Notes:* Gains in both panels use the total-pay baseline mean squared error as the denominator. Approximate worker-clustered standard errors in parentheses condition on fitted forecasts. Panel B compares forecast losses for the same observations within each income group.

with all four components. The additional 0.22 percentage-point gain is statistically significant ($p = 0.025$), while neither the lower- nor middle-income group has a detectable additional gain. Commissions alone leave the higher-income result almost unchanged; the improvement appears when tips and overtime enter together. This comparison does not separate their individual contributions.

The gain within the higher-income group does not establish a difference across income groups. For the all-component specification, the higher–lower difference in forecast gains is not statistically significant ($p = 0.122$). The extension therefore provides evidence that predictive information extends beyond bonuses, while leaving the annual gains small and the income comparison less precisely estimated.

# I Joint Bayesian Learning

The worker updates permanent pay, its changing risk and the fixed parameters from the same payment. This appendix derives the update without treating any of these unknowns as independent in her posterior. Let $\Pi^j_{t-1}$ be her distribution over $(p_{t-1}, q_t, \vartheta)$ conditional on $\mathcal{I}^j_{t-1}$. It describes the possible states and parameters she considers before observing the payment at date $t$.

## I.1 Updating the Probabilities of States and Parameters

Write $\varphi(x; m, v)$ for a Gaussian density with mean $m$ and variance $v$. Conditional on a possible value of $(p_{t-1}, q_t, \vartheta)$, the likelihood of the new payment information is

$$\begin{aligned} L^0_t &= \varphi(w_t; p_{t-1}, q_t + \sigma_u^2 + \sigma_b^2), \\ L^1_t &= \varphi(w_t - b_t; p_{t-1}, q_t + \sigma_u^2)\varphi(b_t; 0, \sigma_b^2). \end{aligned} \tag{I.1}$$

Under $\mathcal{I}^1_t$, the worker observes $b_t$ separately and obtains the remaining pay, $w_t - b_t$. These observations are independent conditional on the states and parameters, so their densities multiply.

Bayes' rule assigns more weight to the possibilities that better explain the payment:

$$d\widetilde{\Pi}_t^j = \frac{L_t^j \, d\Pi_{t-1}^j}{\int L_t^j \, d\Pi_{t-1}^j}. \tag{I.2}$$

This step updates beliefs about last month's permanent pay, current risk and the parameters together. In particular, $\bar{q}$ and $\alpha$ affect the distribution of $q_t$ inherited from earlier payments, so the new observation also provides evidence about them.

### I.2 Inferring the Permanent Shock

For each possible $(p_{t-1}, q_t, \vartheta)$, the remaining pay surprise is $e_t^j = y_t^j - p_{t-1}$, with variance $q_t + R_j$, where $R_0 = \sigma_u^2 + \sigma_b^2$ and $R_1 = \sigma_u^2$. Conditional on these values and the new observation, the permanent shock is Gaussian with mean $\mu_{\psi,t}^j$ and variance $S_{\psi,t}^j$:

$$k_t^j = \frac{q_t}{q_t + R_j}, \qquad \mu_{\psi,t}^j = k_t^j e_t^j, \qquad S_{\psi,t}^j = (1 - k_t^j) q_t. \tag{I.3}$$

To obtain this expression, note that $\psi_t$ and $e_t^j$ are jointly Gaussian conditional on the candidate states and parameters. Their variances are $q_t$ and $q_t + R_j$, and their covariance is $q_t$; the Gaussian conditioning formula therefore gives equation (I.3). The gain $k_t^j$ conditions on a particular value of $p_{t-1}$, unlike $K_t^j$ in the main-text benchmark, which also accounts for uncertainty about that previous level.

The expected squared permanent shock is $S_{\psi,t}^j + (\mu_{\psi,t}^j)^2$. Thus the ARCH equation implies

$$\mathbb{E}[q_{t+1} \mid p_{t-1}, q_t, \vartheta, \mathcal{I}_t^j] = (1 - \alpha)\bar{q} + \alpha \left[ S_{\psi,t}^j + (\mu_{\psi,t}^j)^2 \right]. \tag{I.4}$$

Using only the squared estimated shock would omit $\alpha S_{\psi,t}^j$, understating expected risk whenever the worker remains uncertain about the shock and $\alpha > 0$.

### I.3 Forming the Joint Posterior

For each possibility weighted by $\widetilde{\Pi}_t^j$, the conditional shock distribution and the laws $p_t = p_{t-1} + \psi_t$ and $q_{t+1} = (1 - \alpha)\bar{q} + \alpha\psi_t^2$ determine a distribution over the new states. Integrating these distributions gives $\Pi_t^j$, the joint posterior over $(p_t, q_{t+1}, \vartheta)$. Both new states depend on the same shock, so this step retains their dependence. The resulting posterior need not be Gaussian and becomes the prior for the next payment.

In particular, the permanent-pay moments are

$$\begin{aligned} \widehat{p}_t^j &= \int (p_{t-1} + \mu_{\psi,t}^j) \, d\widetilde{\Pi}_t^j, \\ P_t^j &= \int \left[ S_{\psi,t}^j + (p_{t-1} + \mu_{\psi,t}^j - \widehat{p}_t^j)^2 \right] d\widetilde{\Pi}_t^j. \end{aligned} \tag{I.5}$$

Uncertainty about permanent pay includes both uncertainty within each possibility and disagree-

ment between their implied pay levels. The expected risk state is

$$\widehat{q}^{\,j}_{t+1|t} = \int \left[(1-\alpha)\bar{q} + \alpha\{S^{j}_{\psi,t} + (\mu^{j}_{\psi,t})^2\}\right] d\widetilde{\Pi}^{j}_{t}. \tag{I.6}$$

These expressions retain the posterior dependence between states and parameters. Substituting separate posterior means into the Kalman or ARCH equations generally does not reproduce this joint update.

Observations of $b_t$ inform both the interpretation of the remaining pay and $\sigma_b^2$ itself through the second likelihood factor in equation (I.1). Without this additional information, the temporary variances enter only through their sum, so total pay alone cannot identify their separate values. Beliefs about that split depend on the prior. If the prior links $\sigma_b^2$ to other parameters, observing $b_t$ can also update beliefs about those parameters.

To include recurring payments, let $C \in \{0, 1, \ldots, 12\}$ indicate whether the worker has no annual payment ($C = 0$) or the calendar month in which it arrives. Write $D_t(C)$ for the corresponding payment indicator. For this extension, use

$$\vartheta = (\bar{q}, \alpha, \sigma_u^2, \sigma_v^2, \sigma_a^2, C)$$

in the joint posterior. Conditional on each candidate state, parameter vector and calendar, the payment likelihoods become

$$\begin{aligned} L_t^0 &= \varphi\left(w_t; p_{t-1}, q_t + \sigma_u^2 + \sigma_v^2 + D_t(C)\sigma_a^2\right), \\ L_t^j &= \varphi(w_t - b_t; p_{t-1}, q_t + \sigma_u^2)\varphi\left(b_t; 0, \sigma_v^2 + D_t(C)\sigma_a^2\right), \qquad j = 1, 2. \end{aligned} \tag{I.7}$$

In the conditional shock update, replace $R_j$ by

$$R_{0,t} = \sigma_u^2 + \sigma_v^2 + D_t(C)\sigma_a^2, \qquad R_{1,t} = R_{2,t} = \sigma_u^2,$$

with $y_t^2 = w_t - b_t$. The remaining update proceeds as above, retaining the joint dependence between states, parameters and any unknown calendar. Under $\mathcal{I}_t^0$ and $\mathcal{I}_t^1$, the worker updates the calendar probabilities jointly with the other unknowns. Under $\mathcal{I}_t^2 = \mathcal{I}_t^1 \cup \{C\}$, she conditions on the known calendar. This describes the theoretical extension; Appendix K.1 sets out the numerical treatment.

# J Proofs of the Propositions

### J.1 Proof of Proposition 1

To prove Proposition 1, expand pay at the forecast date as

$$w_{t+12} = p_t + \sum_{h=1}^{12} \psi_{t+h} + u_{t+12} + b_{t+12}. \tag{J.1}$$

Future innovations have mean zero conditional on the parameters and the full history before their arrival. They are therefore uncorrelated with $p_t$ and with earlier innovations, even though their

squares can be dependent. The future temporary components also have zero conditional means and zero covariance with the other terms. Averaging over the worker's uncertainty about states and parameters preserves these zero covariances. Thus

$$V^j_{t,12} = P^j_t + \mathbb{E}[\sigma_u^2 + \sigma_b^2 \mid \mathcal{I}^j_t] + \sum_{h=1}^{12} \mathbb{E}[q_{t+h} \mid \mathcal{I}^j_t]. \tag{J.2}$$

For given $q_{t+1}$ and $\vartheta$, iterating the ARCH conditional mean gives

$$\mathbb{E}[q_{t+h} \mid q_{t+1}, \vartheta, \mathcal{I}^j_t] = \bar{q} + \alpha^{h-1}(q_{t+1} - \bar{q}). \tag{J.3}$$

Summing over $h = 1, \ldots, 12$ and averaging over the joint posterior gives equation (14). This averaging retains dependence between $\alpha$, $\bar{q}$ and $q_{t+1}$. Uncertainty about variance parameters does not add a separate variance-of-variance term: future innovations have the same zero conditional mean for every parameter value, so their contribution is the expected conditional variance.

For recurring payments, substitute $b_{t+12} = v_{t+12} + D_{t+12}a_{t+12}$ in equation (J.1). Future innovations have mean zero conditional on the parameters, the payment schedule and the full history before their arrival, so the same covariance argument applies. The forecast decomposition becomes

$$\begin{aligned} V^j_{t,12} = P^j_t + \mathbb{E}[\sigma_u^2 + \sigma_v^2 + D_{t+12}\sigma_a^2 \mid \mathcal{I}^j_t] + 12\mathbb{E}[\bar{q} \mid \mathcal{I}^j_t] \\ + \mathbb{E}\Big[A_{12}(\alpha)(q_{t+1} - \bar{q}) \mid \mathcal{I}^j_t\Big]. \end{aligned} \tag{J.4}$$

The expectation $\mathbb{E}[D_{t+12}\sigma_a^2 \mid \mathcal{I}^j_t]$ averages jointly over the payment schedule and annual-payment variance. It need not equal $\Pr(D_{t+12} = 1 \mid \mathcal{I}^j_t)\mathbb{E}[\sigma_a^2 \mid \mathcal{I}^j_t]$. Under $\mathcal{I}^2_t$, the worker knows $D_{t+12}$, so it can be taken outside the expectation. The baseline is recovered by setting $D_s = 0$ for every $s$ and $\sigma_v^2 = \sigma_b^2$.

### J.2 Proof of Proposition 2

Proposition 2 follows from the conditional law of total variance and the nesting $\mathcal{I}^0_t \subseteq \mathcal{I}^1_t$:

$$\begin{aligned} \mathrm{Var}(w_{t+12} \mid \mathcal{I}^0_t) = \mathbb{E}[\mathrm{Var}(w_{t+12} \mid \mathcal{I}^1_t) \mid \mathcal{I}^0_t] \\ + \mathrm{Var}\big(\mathbb{E}[w_{t+12} \mid \mathcal{I}^1_t] \mid \mathcal{I}^0_t\big). \end{aligned} \tag{J.5}$$

The future shocks in equation (J.1) have zero conditional mean, so $\mathbb{E}[w_{t+12} \mid \mathcal{I}^1_t] = \widehat{p}^1_t$. Substitution gives the proposition. Strictness requires positive conditional variance of this forecast across the histories of $b_t$ consistent with $\mathcal{I}^0_t$. The comparison averages under the common prior and its implied distribution of histories; it does not establish an ordering of income-group averages evaluated at different fixed true parameters.

The same argument establishes Proposition 2 with recurring payments, since the future innovations remain centred conditional on the parameters and payment schedule. In the comparison between $\mathcal{I}^1_t$ and $\mathcal{I}^2_t$, knowing the schedule similarly removes uncertainty on average when the possible schedules imply different forecasts of permanent pay. The worker may nevertheless report more risk on learning that an annual payment is due in the forecast month, because its

amount remains uncertain.

Under the same common earnings process and prior, the centred-innovation assumptions also give $\mathbb{E}[w_{t+12} \mid \mathcal{I}_t^2] = \widehat{p}_t^2$. Applying the conditional law of total variance to $\mathcal{I}_t^1 \subseteq \mathcal{I}_t^2$ therefore gives

$$\begin{aligned} V_{t,12}^1 - \mathbb{E}[V_{t,12}^2 \mid \mathcal{I}_t^1] &= \mathrm{Var}\big(\mathbb{E}[w_{t+12} \mid \mathcal{I}_t^2] \mid \mathcal{I}_t^1\big) \\ &= \mathrm{Var}(\widehat{p}_t^2 \mid \mathcal{I}_t^1) \geq 0. \end{aligned} \tag{J.6}$$

If knowing the schedule leaves the forecast of permanent pay unchanged, this average reduction is zero.

For the known-variance case with $\alpha = 0$ and no recurring payments, the posterior-variance recursion is $P_t^j = f(P_{t-1}^j, R_j)$, where $f(P, R) = R(P + \bar{q})/(P + \bar{q} + R)$. Its derivatives are

$$\frac{\partial f}{\partial P} = \frac{R^2}{(P + \bar{q} + R)^2} \geq 0, \qquad \frac{\partial f}{\partial R} = \frac{(P + \bar{q})^2}{(P + \bar{q} + R)^2} \geq 0. \tag{J.7}$$

Since $R_1 \leq R_0$, a common initial permanent-pay variance implies $P_t^1 \leq P_t^0$ at every date by induction. All remaining terms in equation (14) are identical under the two information sets, yielding $V_{t,12}^0 - V_{t,12}^1 = P_t^0 - P_t^1 \geq 0$.

The same conditional-variance argument applies to beliefs about long-run permanent risk. Under the common prior and earnings process, and assuming $\mathbb{E}[\bar{q}^2] < \infty$, observing bonuses reduces posterior uncertainty about $\bar{q}$ on average. Iterated expectations imply that its posterior mean is a martingale as information accumulates; these properties do not order the persistence of the worker's full earnings-risk assessments.

### J.3 Proof of Proposition 3

To prove Proposition 3, return to the known-variance benchmark with $\alpha = 0$ and no recurring payments. Before each payment, the new permanent shock adds $\bar{q}$ to uncertainty about permanent pay. Substituting the Kalman gain into $P_t^j = (1 - K_t^j)(P_{t-1}^j + \bar{q})$ gives the recursion $P_t^j = f(P_{t-1}^j, R_j)$, where $f(P, R) = R(P + \bar{q})/(P + \bar{q} + R)$. For $P \geq 0$ and finite $R > 0$, the derivative in equation (J.7) satisfies $0 \leq \partial f/\partial P \leq [R/(\bar{q} + R)]^2 < 1$. The variance therefore converges from any finite initial value to a unique fixed point. At $R = 0$, the payment reveals permanent pay exactly and the posterior variance is zero.

At the fixed point, $P_\infty^j = R_j(P_\infty^j + \bar{q})/(P_\infty^j + \bar{q} + R_j)$, which rearranges to $(P_\infty^j)^2 + \bar{q}P_\infty^j = \bar{q}R_j$. Its non-negative root gives the expression for $P_\infty^j$ in equation (16). Steady state concerns this variance; the worker continues to revise her estimate of permanent pay as new payments arrive. With $\alpha = 0$ and known parameters, the remaining terms in Proposition 1 sum to $12\bar{q} + \sigma_u^2 + \sigma_b^2$, yielding the expression for $V_{\infty,12}^j$.

Holding $\bar{q}$ and total temporary variance $R_0 > 0$ fixed, an increase in $\sigma_b^2$ reduces $R_1 = R_0 - \sigma_b^2$ by the same amount. The worker observing only total pay still faces noise variance $R_0$, while the worker observing $b_t$ separately receives a more precise signal. The contribution of future shocks

to forecast risk remains unchanged, so for $0 < \sigma_b^2 < R_0$,

$$\begin{aligned} \left.\frac{\partial V^1_{\infty,12}}{\partial \sigma_b^2}\right|_{\bar{q},R_0} &= -\frac{\bar{q}}{\sqrt{\bar{q}^{\,2} + 4\bar{q}(R_0 - \sigma_b^2)}} < 0, \\ \left.\frac{\partial V^0_{\infty,12}}{\partial \sigma_b^2}\right|_{\bar{q},R_0} &= 0. \end{aligned} \tag{J.8}$$

The expressions are continuous at the endpoints. Thus a larger separately observed share lowers perceived risk under $\mathcal{I}_t^1$ and increases the reduction relative to total-pay information. The comparison reallocates a fixed amount of temporary variance; increasing $\sigma_b^2$ while holding $\sigma_u^2$ fixed also increases uncertainty about future temporary pay.

With known parameters and a known calendar, the filtering variance can converge to a repeating annual pattern when recurring payments remain in the pay signal. Subtracting the observed $b_t$ leaves constant noise variance $\sigma_u^2$. An unknown calendar is included in the joint posterior described in Appendix I.

### J.4 Proof of Proposition 4

To prove Proposition 4, I first construct the Gaussian approximation without recurring payments, then derive the gain and the carryover of estimation errors. The ARCH transition can be written as

$$q_t = (1-\alpha)\bar{q} + \alpha q_{t-1} + \alpha q_{t-1}(\varepsilon_{t-1}^2 - 1). \tag{J.9}$$

Conditional on $q_{t-1}$ and the parameters, the final term has mean zero and variance $2\alpha^2 q_{t-1}^2$. The Gaussian approximation preserves this conditional mean and replaces the final term by $\eta_t \sim N(0, Q)$. The resulting approximation to permanent-pay risk is

$$h_t = (1-\alpha)\bar{q} + \alpha h_{t-1} + \eta_t.$$

Setting $Q = 2\alpha^2 \bar{q}^{\,2}$ also matches its variance at the long-run risk $\bar{q}$. The result holds for any common $Q > 0$, with $0 < \alpha < 1$.

The pay filter is evaluated at constant reference risk $\bar{q}$, with steady-state variance $P_\infty^j$ from Proposition 3 and temporary noise variance $R_j$. Write $\delta_{t-1}^j = p_{t-1} - \widehat{p}_{t-1}^{\,j}$ for its inherited estimation error and let $r_t^j$ be the temporary noise remaining in $y_t^j$. The pay surprise is $e_t^j = y_t^j - \widehat{p}_{t-1}^{\,j} = \psi_t + \delta_{t-1}^j + r_t^j$. At the reference filter, approximate $\delta_{t-1}^j$ by a mean-zero Gaussian variable with variance $P_\infty^j$, independent of current risk and new shocks. With $\psi_t \mid h_t \sim N(0, h_t)$ and $r_t^j \sim N(0, R_j)$, this gives $e_t^j \mid h_t \approx N(0, h_t + P_\infty^j + R_j)$ locally around positive risk $\bar{q}$.

For a mean-zero Gaussian variable $X$ with variance $v$, $\mathbb{E}[X^2] = v$ and $\mathrm{Var}(X^2) = 2v^2$. Subtracting the reference contributions of inherited estimation error and temporary pay gives the risk reading $\widetilde{h}_t^j = (e_t^j)^2 - P_\infty^j - R_j$, whose conditional moments are

$$\begin{aligned} \mathbb{E}[\widetilde{h}_t^j \mid h_t] &\approx h_t, \\ \mathrm{Var}(\widetilde{h}_t^j \mid h_t) &\approx 2(h_t + P_\infty^j + R_j)^2. \end{aligned} \tag{J.10}$$

Evaluating the second expression at $h_t = \bar{q}$ gives the approximate noise variance

$$\Omega_j = 2(\bar{q} + P_\infty^j + R_j)^2. \tag{J.11}$$

Observing $b_t$ separately lowers both $P_\infty^j$ and $R_j$, making the reading more precise. Squared Gaussian surprises are not themselves Gaussian. Replacing this reading by $s_t^j = h_t + v_t^j$, where $v_t^j \sim N(0, \Omega_j)$, is a further approximation. State innovations and signal errors are independent across months and of each other. This independence is also an approximation: in the ARCH model, the squared permanent shock affects both the pay surprise and next month's risk. The Gaussian process for $h_t$ is local to a positive reference risk; it does not define a globally positive variance process. These assumptions are used for the analytical illustration, while the full model retains the ARCH process and its joint posterior. The reading uses the same payment information as the pay filter and is not an additional independent observation in that joint calculation.

Let $S_t^j = \mathrm{Var}(h_t \mid s_1^j, \ldots, s_t^j)$ in the auxiliary Gaussian model. The worker knows $\bar{q}$, $\alpha$, $Q$ and $\Omega_j$, and the initial state $h_0$ has a Gaussian prior independent of all innovations and signal errors. The usual Kalman recursion gives

$$S_t^{j,-} = \alpha^2 S_{t-1}^j + Q, \qquad K_{h,t}^j = \frac{S_t^{j,-}}{S_t^{j,-} + \Omega_j}, \qquad S_t^j = (1 - K_{h,t}^j) S_t^{j,-}. \tag{J.12}$$

At the steady-state filtering variance, $S_\infty^j = K_h^j \Omega_j$. Substituting this into $S_\infty^j = (1 - K_h^j)(\alpha^2 S_\infty^j + Q)$ yields

$$\alpha^2 \Omega_j (K_h^j)^2 + \left[Q + (1 - \alpha^2)\Omega_j\right] K_h^j - Q = 0. \tag{J.13}$$

The left-hand side is strictly increasing in $K_h^j \geq 0$, negative at zero and positive at one. Its unique root in $(0, 1)$ is

$$K_h^j = \frac{2Q}{Q + (1 - \alpha^2)\Omega_j + \sqrt{\left[Q + (1 - \alpha^2)\Omega_j\right]^2 + 4\alpha^2 \Omega_j Q}}. \tag{J.14}$$

This gives the gain in equation (17). A larger gain increases the response to a given signal surprise. Differentiating equation (J.13) with respect to the signal's noise variance gives

$$\frac{\partial K_h^j}{\partial \Omega_j} = -\frac{\alpha^2 (K_h^j)^2 + (1 - \alpha^2) K_h^j}{2\alpha^2 \Omega_j K_h^j + Q + (1 - \alpha^2)\Omega_j} < 0. \tag{J.15}$$

For the composition comparison, hold $\bar{q}, \alpha, R_0 > 0$ and $Q > 0$ fixed, with $R_1 = R_0 - \sigma_b^2$. Proposition 3 gives $\partial P_\infty^1 / \partial \sigma_b^2 < 0$. Thus

$$\begin{aligned} \frac{\partial \Omega_1}{\partial \sigma_b^2} &= 4(\bar{q} + P_\infty^1 + R_1)\left(\frac{\partial P_\infty^1}{\partial \sigma_b^2} - 1\right) < 0, \\ \frac{\partial K_h^1}{\partial \sigma_b^2} &= \frac{\partial K_h^1}{\partial \Omega_1} \frac{\partial \Omega_1}{\partial \sigma_b^2} > 0. \end{aligned} \tag{J.16}$$

These derivatives hold the stated quantities fixed. A positive variance $\sigma_b^2$ also implies $R_1 < R_0$ and

$P^1_\infty < P^0_\infty$, so $\Omega_1 < \Omega_0$ and $K^1_h > K^0_h$.

Finally, let $\xi^j_t = \widehat{h}^j_t - h_t$ denote the estimation error. Subtracting the state transition from the belief update gives

$$\xi^j_t = \alpha(1 - K^j_h)\xi^j_{t-1} - (1 - K^j_h)\eta_t + K^j_h \nu^j_t. \tag{J.17}$$

The previous error therefore carries into the next error with coefficient $\alpha(1 - K^j_h)$. Holding the subsequent state innovations and signal errors fixed, the contribution of an error $k$ months earlier is multiplied by $[\alpha(1 - K^j_h)]^k$, which declines faster at a larger gain. For the estimated risk level itself, however, $\mathbb{E}[\widehat{h}^j_t - \bar{q} \mid s^j_1, \ldots, s^j_{t-1}] = \alpha(\widehat{h}^j_{t-1} - \bar{q})$ under either information set in the Gaussian model. The result on error correction is therefore distinct from a result on the persistence of reported risk.

In the exact ARCH model with known parameters, iterated expectations give

$$\begin{aligned} \mathbb{E}[\widehat{q}^j_{t+2|t+1} - \bar{q} \mid \mathcal{I}^j_t] &= \mathbb{E}[q_{t+2} - \bar{q} \mid \mathcal{I}^j_t] \\ &= \alpha\big(\widehat{q}^j_{t+1|t} - \bar{q}\big). \end{aligned} \tag{J.18}$$

The second equality uses $\mathbb{E}[\psi^2_{t+1} \mid q_{t+1}, \vartheta, \mathcal{I}^j_t] = q_{t+1}$. This conditional-mean relation holds under either information set. It concerns the worker's expected permanent-shock variance, rather than the persistence coefficient of her full earnings-risk report.

# K Quantitative Implementation and Sensitivity

### K.1 Numerical Learning

Workers begin with prior distributions common across income groups. Under $\mathcal{I}^0$ and $\mathcal{I}^1$, let $\sigma^2_T = \sigma^2_u + \sigma^2_b$. The variance priors are

$$\begin{aligned} \log \bar{q} &\sim N\Big(\log \bar{q}_0 - \tfrac{1}{2}\tau^2_q, \tau^2_q\Big), \\ \log \sigma^2_T &\sim N\Big(\log \sigma^2_{T,0} - \tfrac{1}{2}\tau^2_T, \tau^2_T\Big). \end{aligned} \tag{K.1}$$

I set $\bar{q}_0 = 8.65$ and $\sigma^2_{T,0} = 3.06$ from the SIPP cross-group averages, and maintain $\tau_q = \tau_T = 2.5$. An independent share $s \sim U(0,1)$ assigns $\sigma^2_b = s\sigma^2_T$ and $\sigma^2_u = (1 - s)\sigma^2_T$. Under $\mathcal{I}^2$, the same prior family and means apply to $\sigma^2_u + \sigma^2_v$ and its irregular-bonus share. A separate lognormal prior for annual-payment variance has mean 40.7 and log standard deviation 1.5. Workers also learn $\alpha$, starting from $U(0, 0.55)$. The prior distributions remain maintained assumptions; their parameters are not fitted to the SCE.

I approximate each worker's distribution over permanent-pay histories and unknown parameters with a resample–move particle filter. The main runs use 1,024 particles initialised at scrambled Sobol points of the common prior. At each payment, a particle supplies a candidate past permanent-pay path, current risk and parameter vector. Conditional on these candidates, the Gaussian signal update gives a proposal for current permanent pay. Its predictive likelihood determines the particle's weight. Under $\mathcal{I}^1$, the likelihood also includes the separately observed bonus. Under $\mathcal{I}^2$, the bonus likelihood for irregular variance excludes annual-payment months;

the bonus is still subtracted from total pay in those months when filtering permanent pay.

I calculate the report from the weighted conditional moments before systematic resampling. The forecast contribution of permanent shocks is

$$\sum_{h=1}^{12} \mathbb{E}[q_{t+h} \mid \mathcal{I}_t^j] = \mathbb{E}\Big[12\bar{q} + A_{12}(\alpha)(q_{t+1} - \bar{q}) \mid \mathcal{I}_t^j\Big]. \tag{K.2}$$

The expectation is taken jointly over states and parameters. In particular, I do not replace the product involving $\alpha$ and current risk with a product of posterior means. After resampling, four Metropolis–Hastings moves update the parameters and permanent-pay path. Path proposals use forward filtering and backward sampling under a constant-variance reference process. The acceptance probability corrects for ARCH risk. Under $\mathcal{I}^0$, the move changes total temporary variance and redraws its unidentified split from the conditional prior.

Initial permanent pay has prior $N(0, 10{,}000)$. True earnings histories start risk from 256 ARCH burn-in draws. The filter approximates its stationary initial-risk distribution using 48 auxiliary draws per particle, which are also updated in the rejuvenation step. These initialisations distinguish the simulated process from the worker's uncertainty about its starting state.

Annual-payment variance is updated outside the particle filter. Let $z = \log \sigma_a^2$ and write $\widehat{\sigma}_{v,t}^2 = \mathbb{E}[\sigma_v^2 \mid \mathcal{I}_t^2]$. At a scheduled payment month, I evaluate the density on 4,000 equally spaced points between $\log(10^{-3})$ and $\log(10^5)$:

$$\pi_t(z) \;\propto\; \exp\left[-\frac{(z-\mu_a)^2}{2\tau_a^2}\right] \prod_{\substack{s \le t \\ D_s = 1}} \varphi\left(b_s; 0, \widehat{\sigma}_{v,t}^2 + e^z\right), \qquad \mu_a = \log(40.7) - \frac{1.5^2}{2}, \quad \tau_a = 1.5, \tag{K.3}$$

where $\varphi(\cdot; m, v)$ denotes the Gaussian density. The implementation uses the unrounded cross-group mean for the prior centre. Normalised grid weights give the posterior mean of $e^z$. The report includes that mean only when $D_{t+12} = 1$, which coincides with $D_t = 1$ for the fixed annual schedule. Holding irregular variance at its current mean and excluding annual months from its bonus likelihood makes this an approximation to joint inference about the two variances.

Without calendar information, the worker includes her estimated bonus variance in every forecast: $B_{t,12}^j = \mathbb{E}[\sigma_b^2 \mid \mathcal{I}_t^j]$. With calendar information, I use equation (13), which includes annual-payment uncertainty only when the worker forecasts her scheduled month. This timing effect relies on interpreting the SCE report as uncertainty about earnings in month $t + 12$. A forecast of total earnings over the coming twelve months would include an annual payment in every window.

Under $\mathcal{I}^0$ and $\mathcal{I}^1$, workers estimate monthly temporary variance from their payments but do not learn the payment calendar. The comparison with $\mathcal{I}^2$ therefore changes both the available information and the forecasting model. The information comparison in Appendix J.2 assumes a common earnings process and prior, so it does not automatically apply to this numerical calendar comparison.

### K.2 Calibration Checks

For each income group, the variance decomposition in Section 2 sets long-run permanent variance to $\bar{q}_g = \mathcal{P}_g$ and total temporary variance to $\mathcal{T}_g$. The component decomposition in Section 3.1 assigns the bonus contribution to $\sigma^2_{b,g}$, leaving $\sigma^2_{u,g} = \mathcal{T}_g - \sigma^2_{b,g}$. All inputs use the same-job SIPP sample allowing reported hours to change.

A share $\rho$ of workers has an annual payment in a fixed calendar month, distributed uniformly across these workers. Averaging across workers and calendar months gives

$$\sigma^2_v + \frac{\rho}{12}\sigma^2_a = \sigma^2_b. \tag{K.4}$$

I impose this identity when matching the empirical bonus contribution, so the recurring payment accounts for part of the same temporary variation.

Bonus timing is measured among workers with enough same-job history to observe the corresponding month one year earlier. Let $\pi_g$ be the share of months with bonus receipt one year earlier, and let $B^{\text{yes}}_g$ and $B^{\text{no}}_g$ denote the bonus variance contributions in those months and the remaining months. Writing $B^{\text{lb}}_g$ for the average contribution in this lookback sample, the mapping is

$$\rho_g = 12\pi_g, \qquad \sigma^2_{a,g} = \kappa_g(B^{\text{yes}}_g - B^{\text{no}}_g), \qquad \sigma^2_{v,g} = \kappa_g B^{\text{no}}_g, \qquad \kappa_g = \frac{\sigma^2_{b,g}}{B^{\text{lb}}_g}. \tag{K.5}$$

One annual month for a share $\rho_g$ of workers gives an annual-payment frequency of $\rho_g/12$. The excess variation in recurring months determines the annual variance. Scaling both components by $\kappa_g$ reconciles the lookback sample with the full calibration sample and satisfies equation (K.4). This mapping approximates observed recurrence by a fixed annual schedule; it does not establish that workers know that schedule.

The ARCH coefficient $\alpha$ determines how quickly elevated permanent-shock risk decays in expectation. The SIPP volatility proxy also reflects temporary payments, whose arrival and disappearance can create consecutive large earnings changes. Its persistence coefficient therefore need not equal $\alpha$.

The ARCH calibration simulates 2,048 workers per income group for 47 months at each candidate $\alpha \in \{0.30, 0.32, \ldots, 0.50\}$. I pool earnings changes across groups when removing their linear dependence on the previous change, then apply the main system GMM specification to twelve-month windows of squared residuals. Table K.1 shows the fit around the selected value. The objective weights each squared group discrepancy by the inverse squared SIPP standard error. It is relatively flat between 0.42 and 0.46. The selected value approximates overall persistence without reproducing its income gradient.

The simulation uses the empirical proxy construction and instrument specification, but does not reproduce the survey panel exactly. Model windows are balanced, equally weighted and untrimmed, whereas the SIPP uses survey weights, trimming and job spells of varying length. The Gaussian innovation specification with ARCH risk also permits large earnings movements. The fitted coefficients should therefore be read as a calibration to the empirical proxy, with these sampling differences.

**Table K.1:** Calibration to earnings-risk persistence

| | Lower | Middle | Higher | Weighted distance |
|---|---|---|---|---|
| SIPP | 0.193 | 0.187 | 0.235 | – |
| *Simulated proxy by $\alpha$* | | | | |
| 0.30 | 0.138 | 0.161 | 0.169 | 77.8 |
| 0.42 | 0.195 | 0.200 | 0.200 | 16.4 |
| 0.44 | 0.206 | 0.207 | 0.206 | 15.2 |
| 0.46 | 0.217 | 0.215 | 0.212 | 17.6 |

*Notes:* The grid uses the earlier three-group earnings process; workers' beliefs do not enter. SIPP standard errors are 0.013, 0.010 and 0.009. The income-dependent calibration retains $\alpha = 0.44$ (Appendix K.6).

### K.3 Learning Histories and Persistence

I simulate 8,192 workers per job-start income stratum and weight the strata to their population shares. Results use groups defined by earnings over the twelve months ending at the first report and drop workers with any report outside the pooled first and ninety-ninth percentiles, separately for each information set. Workers update their beliefs from the first payment. For each learning history $A$, I retain twelve reports after payments $A, \ldots, A + 11$ from the same simulated histories. The benchmark uses $A = 48$, and the learning figures extend the comparison through $A = 72$, requiring reports through payment 83. The history is common across income groups and represents available payment information, rather than measured employment tenure.

Table K.2 summarises the benchmark calibration under all three information sets, reporting the fit to persistence under $\mathcal{I}^2$, predictions under the other information sets and mean perceived risk relative to higher-income workers.[15]

Figure K.1 shows the calendar effect by aligning annual-payment recipients around their scheduled month. With payment timing known, risk rises briefly when the forecast concerns an annual payment and falls back in the following month. The annual-payment contribution in that month averages 28, 51 and 57 variance units across the three income groups. These paths concern recipients; the population effect also depends on how many workers receive annual payments.

To isolate variation in the annual-payment contribution, I also construct a diagnostic series from the $\mathcal{I}^2$ reports at $A = 48$. For each worker, I subtract the scheduled-payment component from each of her twelve reports and add back its average over those reports. Her mean report, posterior beliefs and all remaining components are unchanged. I retain the workers selected by trimming the original reports, their income groups, population weights and the same resampling indices used for GMM. Persistence in the resulting series is 0.299, 0.359 and 0.362, compared with 0.282, 0.244 and 0.179 in the original series. This calculation removes temporal variation in the annual-payment component; it does not re-solve the worker's learning problem under a different information set.

Figure K.2 shows the mean-risk levels underlying the ratios in Figure 5. Each point is the group mean immediately after payment $A$.

[15] To assess numerical precision, I repeat inference on matched workers with twice as many particles (Appendix K.5).

**Table K.2:** Perceived risk by payment information

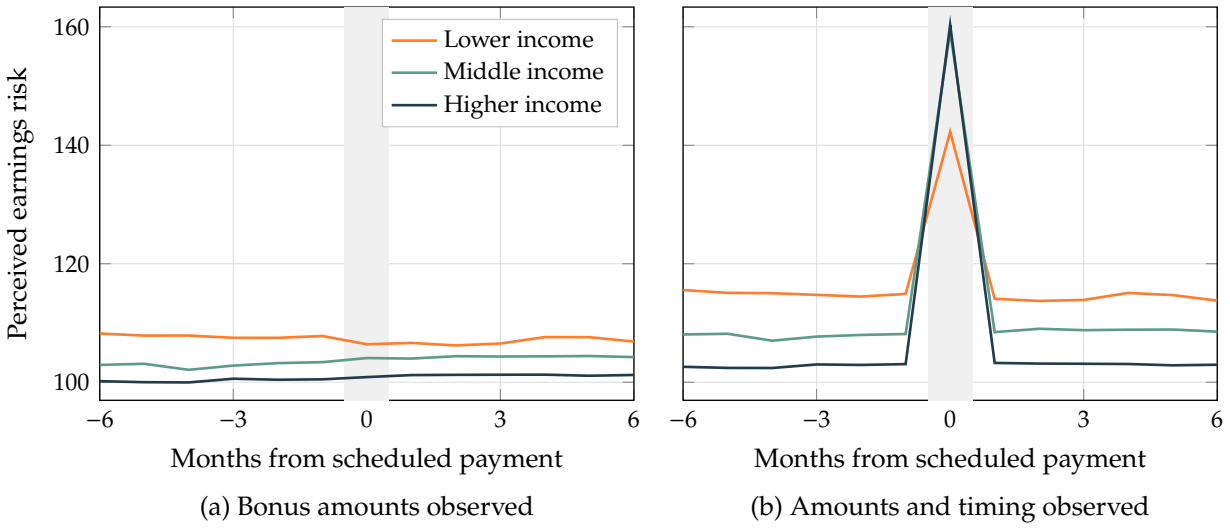
*A. Mean risk relative to higher income*

| | Lower/higher | Middle/higher |
|---|---|---|
| SCE | 1.81 (0.130) | 0.99 (0.064) |
| Total pay, $\mathcal{I}^0$ | 0.95 (0.005) | 0.97 (0.005) |
| Bonus amounts, $\mathcal{I}^1$ | 1.11 (0.005) | 1.05 (0.005) |
| Amounts and timing, $\mathcal{I}^2$ | 1.11 (0.005) | 1.05 (0.005) |

*B. Monthly persistence*

| | Lower | Middle | Higher | Lower−higher |
|---|---|---|---|---|
| SCE | 0.343 (0.046) | 0.142 (0.026) | 0.144 (0.025) | 0.199 |
| Total pay, $\mathcal{I}^0$ | 0.401 (0.016) | 0.392 (0.016) | 0.445 (0.014) | -0.044 (0.021) |
| Bonus amounts, $\mathcal{I}^1$ | 0.310 (0.015) | 0.351 (0.016) | 0.346 (0.014) | -0.035 (0.020) |
| Amounts and timing, $\mathcal{I}^2$ | 0.282 (0.014) | 0.244 (0.014) | 0.179 (0.012) | 0.104 (0.020) |

*Notes:* Model estimates use population weights and income groups defined by earnings over the twelve months ending at the first assessment. Trimming excludes workers with any assessment outside the pooled first and ninety-ninth percentiles, separately by information set. Model standard errors in parentheses use 100 worker-bootstrap draws conditional on the calibration and maintained assumptions. SCE inference follows Appendix D.3.

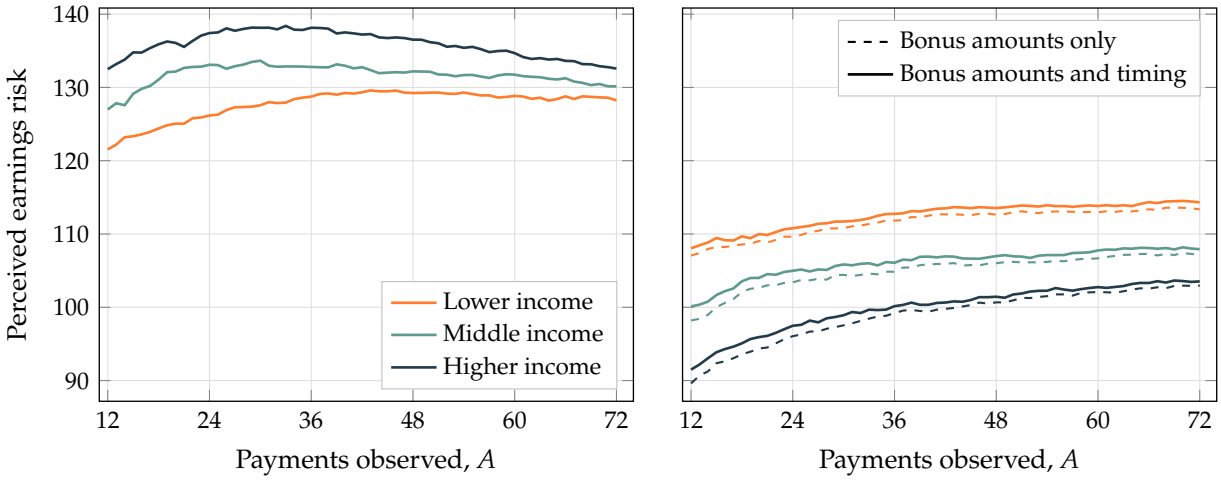



(a) Bonus amounts observed

(b) Amounts and timing observed

**Figure K.1:** Perceived risk around a scheduled annual payment

*Notes:* Mean twelve-month forecast variance among 3,329 annual-payment recipients, in squared 100-log-point units. Month zero (shaded) aligns each worker's scheduled payment between payments 42 and 53. Panel (a) uses bonus amounts alone ($\mathcal{I}^1$); panel (b) adds payment timing ($\mathcal{I}^2$). Both panels use the same workers, grouped by earnings over the twelve payments before month zero. Means use population weights, without trimming or smoothing.




(a) Total pay only (b) Bonus pay observed

**Figure K.2:** Perceived earnings risk by learning history

*Notes:* Mean perceived risk immediately after payment $A$, in squared 100-log-point units. Grouping, weights and trimming follow Figure 4. Points are unsmoothed. At $A = 48$, each worker contributes one assessment; Table K.2 instead averages assessments after payments 48–59.

### K.4 Posterior Beliefs and Prior Sensitivity

Table K.3 reports mean beliefs over the twelve reports at $A = 48$. Bonus information reverses the income ordering of estimated long-run permanent risk. Calendar information changes these estimates little. The posterior mean of $\alpha$ stays close to its prior mean of 0.275 under all three information sets, so broad priors on variance levels do not imply that all parameters have been learned accurately.

**Table K.3:** Posterior beliefs about permanent risk

| | $\mathbb{E}[\bar{q} \mid \mathcal{I}_t^j]$ | | | $\mathbb{E}[\alpha \mid \mathcal{I}_t^j]$ | | |
|---|---|---|---|---|---|---|
| | Lower | Middle | Higher | Lower | Middle | Higher |
| SIPP group target | 8.93 | 8.64 | 8.39 | 0.440 | 0.440 | 0.440 |
| Total pay, $\mathcal{I}^0$ | 10.57 | 10.67 | 10.92 | 0.279 | 0.276 | 0.276 |
| Bonus amounts, $\mathcal{I}^1$ | 9.19 | 8.51 | 7.97 | 0.286 | 0.288 | 0.293 |
| Amounts and timing, $\mathcal{I}^2$ | 9.27 | 8.59 | 8.06 | 0.286 | 0.288 | 0.292 |

*Notes:* Mean posterior beliefs after payments 48–59, using the groups, weights and trimming in Table K.2. Information sets share earnings histories and priors are common across income groups. The first row gives SIPP targets for group averages at job entry. Permanent variances are in squared 100-log-point units.

Supporting comparisons under $\alpha = 0.30$ assess the role of the prior. A grid of common log standard deviations from 0.5 to 2.5 for permanent variance and from 0.25 to 2.5 for monthly temporary variance leaves the lower–higher level ratio between approximately 0.89 and 1.18 across the evaluated settings. This search used 512 workers per group; its persistence estimates are too noisy to rank nearby specifications. An extension of this earlier fixed-type exercise considers log standard deviations of up to 5. Across learning ages 12, 24 and 36, the lower–higher ratio

remains below 1.2. The prior means are held fixed, so widening these lognormal priors also lowers their medians. In an earlier monthly-bonus model with $\alpha = 0.30$, independent priors on bonus and other temporary variance gave a level ratio of about 1.03 with observed bonuses, compared with 1.13 under the total-and-share prior. These exercises indicate sensitivity of the recognition effect to prior dependence without explaining the SCE ratio of 1.81. They do not constitute a prior-sensitivity exercise at the main calibration $\alpha = 0.44$.

### K.5 Numerical Precision

Simulation standard errors use 100 bootstrap samples of workers within each income group, conditional on the earnings parameters, priors and benchmark learning history.

The main results pool sixteen batches of 512 workers per job-start income stratum. I repeat inference for 768 workers drawn across these batches, using 2,048 particles and eight Metropolis–Hastings moves in place of 1,024 particles and four moves. The earnings histories, filter seeds, income groups and retained samples remain the same. Mean perceived risk changes by less than 0.4 percent in every income group and information set. Persistence differences reach 0.059 under $\mathcal{I}^1$ and 0.040 under $\mathcal{I}^2$. Paired worker-bootstrap comparisons remain imprecise, so this check does not establish that numerical uncertainty is negligible in every persistence estimate.

### K.6 Continuous Income and Smooth Parameter Schedules

I use anchor incomes of \$35,100, \$73,788 and \$158,796 and choose the parameter values at these anchors so that their averages within the three job-start income groups reproduce the SIPP targets. Between the anchors, I interpolate the logarithms of $\bar{q}$, $\sigma_b^2$, $\sigma_u^2$, $\rho$ and $\sigma_a^2$ using a shape-preserving cubic in log income. Outside this range, each log schedule continues with a bounded hyperbolic-tangent tail that preserves its endpoint slope, using a scale of 0.5 log-income units. This continuation is a maintained extrapolation assumption. Starting log annual income follows a normal distribution with mean 11.48849 and standard deviation 0.754. I draw 8,192 workers within each income stratum and weight them by the implied population shares.

In month $t$, other temporary variance, irregular-bonus variance and annual-payment variance follow these schedules at permanent income entering the month. Irregular-bonus variance is given by the schedule $\sigma_v^2(x) = \sigma_b^2(x) - \rho(x)\sigma_a^2(x)/12$. Long-run permanent variance, annual-payment eligibility and the payment month remain fixed at job entry, and $\alpha = 0.44$ is common. Since eligibility remains fixed, the initial bonus-variance identity need not continue to hold at each worker's subsequent income. Crossing an income threshold changes the worker's possible classification without a discrete change in her temporary-pay variances.

Workers estimate unknown, constant temporary-variance parameters from their own payments. They do not use income to infer these parameters or estimate how they vary with income. Their posterior assessments can change after each payment, but their subjective model does not include the changing variance schedules used to generate earnings. This is a maintained approximation to learning about the evolving process.

For the twelve-month forecast, I average the temporary-variance parameters over their posterior distribution. Under $\mathcal{I}^2$, annual-payment variance is learned separately using equation (K.3). The

permanent-risk forecast remains as in equation (K.2).

A check of the calibrated earnings process on identical simulated histories finds that allowing temporary variances to change with income alters the group earnings-variance moments by at most 2.3 percent relative to fixed composition. Permanent-variance estimates change by at most 0.01, and the volatility-proxy persistence estimates remain similar. I therefore retain the calibrated schedules and $\alpha = 0.44$.

The main results classify workers by realised earnings over the twelve months ending at the first report and hold this classification fixed over the reporting window. This follows the SCE's entry-income grouping, approximating household income with the worker's own earnings. Workers can change group across learning histories. In the temporary-payment experiment, groups use earnings over the twelve months before the intervention and remain fixed thereafter. In the policy experiment, I update groups each month using earnings over the latest twelve payments, excluding transfers. The income thresholds remain unchanged, and each comparison uses the same workers in the transfer and no-transfer cases at a given date.